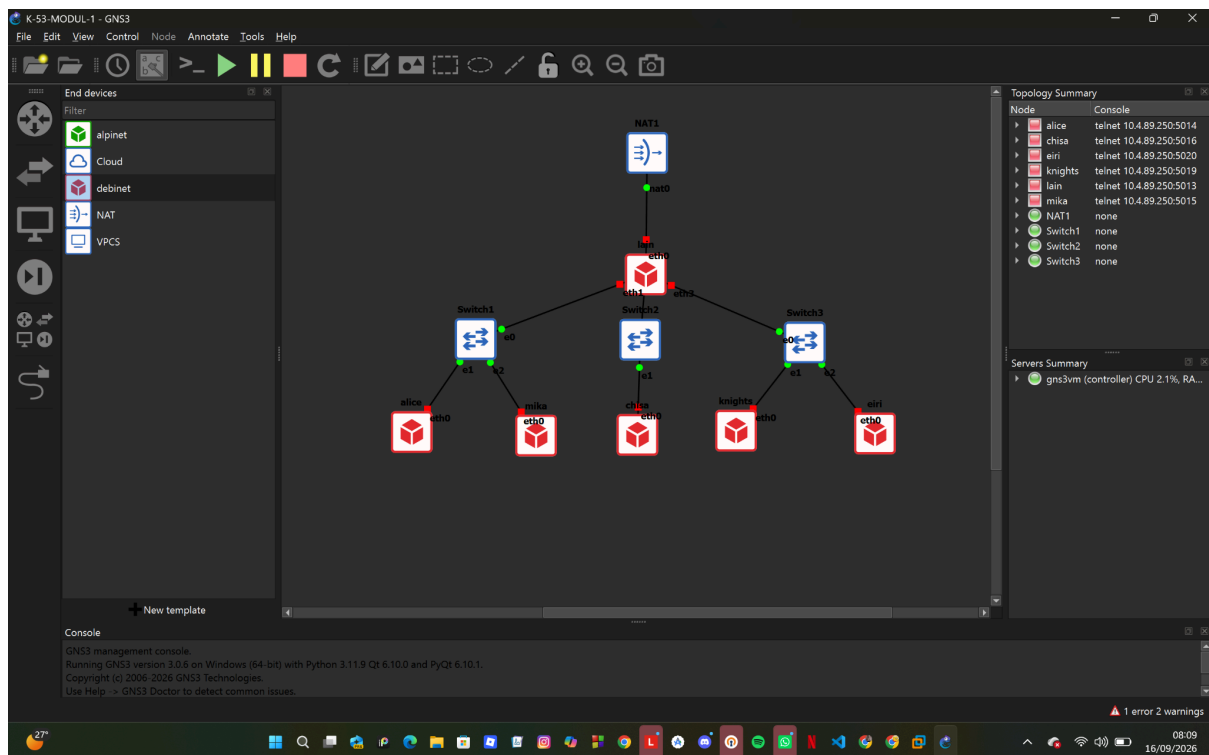


DOKUMENTASI PRAKTIKUM

KOMDAT JARKOM

K-053

1. Untuk mempersiapkan pembangunan The Wired, Lain yang berperan sebagai Router membuat tiga Switch/Gateway: Switch 1 menuju dua Entitas yaitu Alice dan Mika, Switch 2 menuju Chisa, sedangkan Switch 3 menuju Knights dan Eiri. Kelima Entitas tersebut dikonfigurasi sebagai Client di GNS3. **[GUNAKAN PREFIX IP MASING-MASING KELOMPOK]**



```
lain
auto eth0
iface eth0 inet dhcp
auto eth1
iface eth1 inet static
    address 10.190.1.1
    netmask 255.255.255.0
auto eth2
iface eth2 inet static
    address 10.190.2.1
    netmask 255.255.255.0
auto eth3
iface eth3 inet static
```

```
address 10.190.3.1
```

```
netmask 255.255.255.0
```

alice

```
auto eth0
```

```
iface eth0 inet static
```

```
    address 10.190.1.2
```

```
    netmask 255.255.255.0
```

```
    gateway 10.190.1.1
```

mika

```
auto eth0
```

```
iface eth0 inet static
```

```
    address 10.190.1.3
```

```
    netmask 255.255.255.0
```

```
    gateway 10.190.1.1
```

chisa

```
auto eth0
```

```
iface eth0 inet static
```

```
    address 10.190.2.2
```

```
    netmask 255.255.255.0
```

```
    gateway 10.190.2.1
```

knights

```
auto eth0
```

```
iface eth0 inet static
```

```
    address 10.190.3.2
```

```
    netmask 255.255.255.0
```

```
    gateway 10.190.3.1
```

eiri

```
auto eth0
```

```
iface eth0 inet static
```

```
    address 10.190.3.3
```

```
    netmask 255.255.255.0
```

```
    gateway 10.190.3.1
```

2. Karena menurut Lain pada saat itu The Wired masih terisolasi dari dunia luar, konfigurasi router Lain agar dapat tersambung langsung ke jaringan internet publik melalui NAT/DHCP pada interface eth0

```
echo "nameserver 8.8.8.8" > /etc/resolv.conf
sysctl -w net.ipv4.ip_forward=1
```

verifikasi

```
ping -c 2 google.com
```

```
root@lain:~# echo "nameserver 8.8.8.8" > /etc/resolv.conf
root@lain:~# sysctl -w net.ipv4.ip_forward=1
net.ipv4.ip_forward = 1
root@lain:~# ping -c 2 google.com
PING google.com (64.233.170.101) 56(84) bytes of data.
64 bytes from sg-in-f101.1e100.net (64.233.170.101): icmp_seq=1 ttl=102 time=22.
6 ms
64 bytes from sg-in-f101.1e100.net (64.233.170.101): icmp_seq=2 ttl=102 time=22.
0 ms

--- google.com ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1001ms
rtt min/avg/max/mdev = 21.985/22.290/22.596/0.305 ms
root@lain:~#
```

3. Setelah router Lain terhubung ke internet, pastikan seluruh Entitas (Client) di bawah Switch 1, Switch 2, dan Switch 3 dapat saling terhubung dan berkomunikasi satu sama lain melalui konfigurasi routing.

aktifkan IP Forwarding di router **Lain**:

```
sysctl -w net.ipv4.ip_forward=1
```

pembuatan

Dari Alice (Subnet 1) ke Chisa (Subnet 2):

```
ping -c 2 10.190.2.2
```

```
root@alice:~# ping -c 2 10.190.2.2
PING 10.190.2.2 (10.190.2.2) 56(84) bytes of data.
64 bytes from 10.190.2.2: icmp_seq=1 ttl=63 time=0.695 ms
64 bytes from 10.190.2.2: icmp_seq=2 ttl=63 time=0.445 ms

--- 10.190.2.2 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1060ms
rtt min/avg/max/mdev = 0.445/0.570/0.695/0.125 ms
root@alice:~#
```

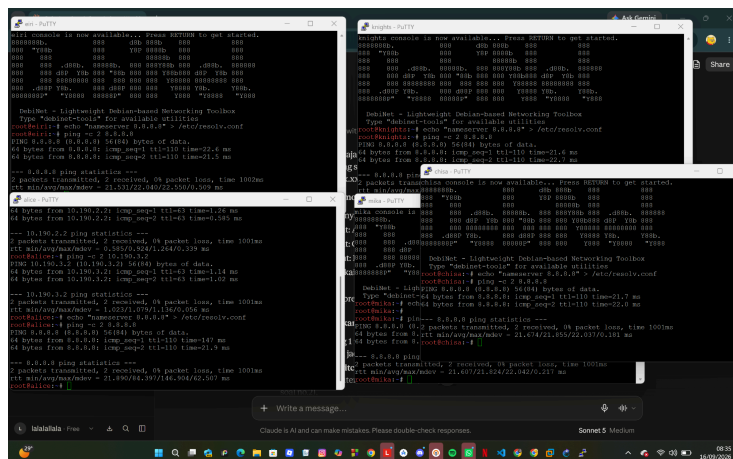
Dari Knights (Subnet 3) ke Mika (Subnet 1):

ping -c 2 10.190.1.3

```
root@knights:/# ping -c 2 10.190.1.3
PING 10.190.1.3 (10.190.1.3) 56(84) bytes of data.
64 bytes from 10.190.1.3: icmp_seq=1 ttl=63 time=0.829 ms
64 bytes from 10.190.1.3: icmp_seq=2 ttl=63 time=0.509 ms

--- 10.190.1.3 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1052ms
rtt min/avg/max/mdev = 0.509/0.669/0.829/0.160 ms
[1]+  Done                  tshark -i eth0 -f "icmp" -w /tmp/soal10.pcap > /dev/null 2>&1
root@knights:/#
```

4. Lain ingin agar setiap Entitas (Client) memiliki kemandirian di The Wired. Konfigurasi firewall/iptables (NAT Masquerade) dan DNS resolver agar setiap Client dapat terhubung ke internet secara mandiri (dapat melakukan ping ke 8.8.8.8 dan membuka domain web google.com).



```
sysctl -w net.ipv4.ip_forward=1
```

```
iptables -t nat -A POSTROUTING -o eth0 -j MASQUERADE
iptables -A FORWARD -i eth1 -o eth0 -j ACCEPT
iptables -A FORWARD -i eth2 -o eth0 -j ACCEPT
iptables -A FORWARD -i eth3 -o eth0 -j ACCEPT
iptables -A FORWARD -i eth0 -m state --state ESTABLISHED,RELATED -j ACCEPT
```

echo "nameserver 8.8.8.8" > /etc/resolv.conf ke masing masing

ceknya pake

ping -c 2 8.8.8.8

ping -c 2 [google.com](https://www.google.com)

5. Eiri tetap berupaya menanamkan kekacauan ke dalam jaringan. Untuk mengantisipasi restart tiba-tiba, pastikan seluruh konfigurasi jaringan tidak hilang saat semua node di-restart. Buat script verifikasi di `/root/cek_status.sh` pada router Lain yang menampilkan ringkasan interface (`ip -br a`) dan status tabel NAT (`iptables -t nat -L -v -n`) setelah reboot.

manambahkan config dilain

```
auto eth0
iface eth0 inet dhcp
auto eth1
iface eth1 inet static
    address 10.190.1.1
    netmask 255.255.255.0
auto eth2
iface eth2 inet static
    address 10.190.2.1
    netmask 255.255.255.0
auto eth3
iface eth3 inet static
    address 10.190.3.1
    netmask 255.255.255.0
up sysctl -w net.ipv4.ip_forward=1
up iptables -t nat -A POSTROUTING -o eth0 -j MASQUERADE
```

terus

`nano /root/cek_status.sh`

```
#!/bin/sh
```

```
echo "=====
echo "    RINGKASAN INTERFACE IP    "
echo "=====
ip -br a

echo ""
echo "=====
echo "    STATUS TABEL NAT (IPTABLES)    "
echo "=====
iptables -t nat -L -v -n
```

```
chmod +x /root/cek_status.sh
/root/cek_status.sh
```

```
root@lain:~# chmod +x /root/cek_status.sh
/root/cek_status.sh
=====
RINGKASAN INTERFACE IP
=====
lo                UNKNOWN      127.0.0.1/8  ::1/128
eth0              UNKNOWN      192.168.122.175/24
eth1              UNKNOWN      10.190.1.1/24 fe80::42:b0ff:fe1:f101/64
eth2              UNKNOWN      10.190.2.1/24 fe80::42:b0ff:fe1:f102/64
eth3              UNKNOWN      10.190.3.1/24 fe80::42:b0ff:fe1:f103/64
=====
STATUS TABEL NAT (IPTABLES)
=====
Chain PREROUTING (policy ACCEPT 0 packets, 0 bytes)
  pkts bytes target      prot opt in     out     source destination
Chain INPUT (policy ACCEPT 0 packets, 0 bytes)
  pkts bytes target      prot opt in     out     source destination
Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
  pkts bytes target      prot opt in     out     source destination
Chain POSTROUTING (policy ACCEPT 0 packets, 0 bytes)
  pkts bytes target      prot opt in     out     source destination
      0      0 MASQUERADE all  --  *      eth0    0.0.0.0/0  0.0.0.0/0
```

6. Mika mencurigai adanya anomali traffic pada segmen jaringannya. Jalankan generator traffic berikut ([link file](#)) pada node Mika, lalu lakukan packet sniffing menggunakan Wireshark pada interface node Mika. Terapkan display filter khusus untuk menyaring paket yang berprotokol DNS atau ICMP. Tunjukkan screenshot hasil filter beserta ringkasan paket yang lolos.

```
nano /root/traffic_protocol7.sh
```

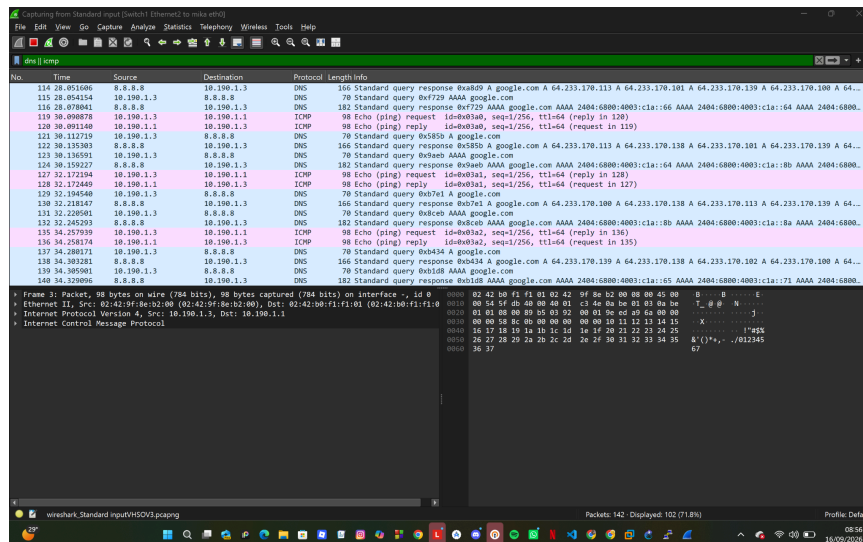
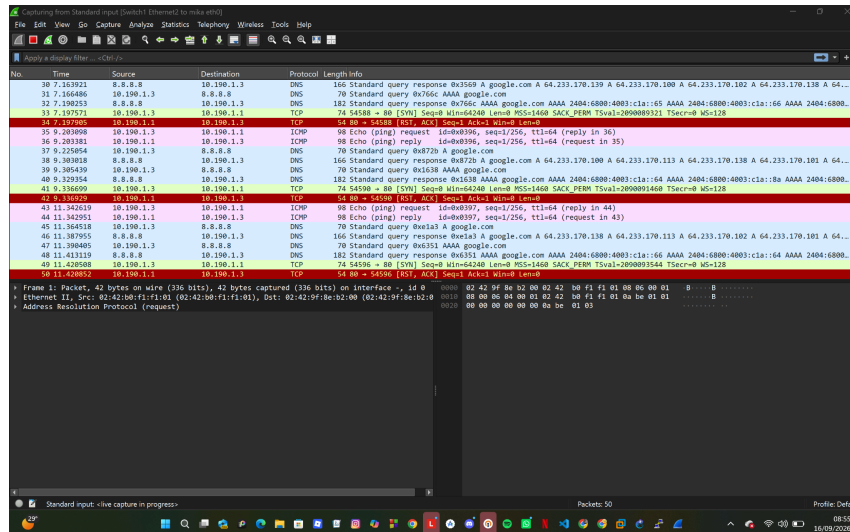
```
#!/bin/sh

while true; do
  ping -c 1 10.190.1.1 > /dev/null 2>&1
  nslookup google.com 8.8.8.8 > /dev/null 2>&1
  nc -z -w 1 10.190.1.1 80 > /dev/null 2>&1
```

sleep 2
done

chmod +x /root/traffic_protocol7.sh
/root/traffic_protocol7.sh &

buka wireshark mika terus dns || icmp



7.1. Chisa memutuskan mendirikan FTP Server pada node miliknya dengan shared folder di /var/wired/data. Terapkan kebijakan akses: user alice (hak akses read & write), user mika (dibatasi read-only), dan user eiri (dibatasi tanpa izin akses / blacklist). Buktikan konfigurasi dengan membuat file *signal_alice.txt* dari user alice, dan buktikan penolakan akses saat user eiri mencoba login.

dichisa

```
echo "nameserver 8.8.8.8" > /etc/resolv.conf  
apt update && apt install -y --allow-unauthenticated vsftpd
```

```
mkdir -p /var/wired/data  
chmod 777 /var/wired/data  
chown -R ftp:nogroup /var/wired/data
```

```
useradd -m -s /bin/ alice 2>/dev/null  
useradd -m -s /bin/ mika 2>/dev/null  
useradd -m -s /bin/ eiri 2>/dev/null
```

```
echo "alice:password" | chpasswd  
echo "mika:password" | chpasswd  
echo "eiri:password" | chpasswd
```

```
cat << 'EOF' > /etc/vsftpd.conf  
listen=YES  
listen_ipv6=NO  
anonymous_enable=NO  
local_enable=YES  
write_enable=YES  
local_umask=022  
check_shell=NO
```

```
local_root=/var/wired/data  
chroot_local_user=YES  
allow_writeable_chroot=YES
```

```
userlist_enable=YES  
userlist_file=/etc/vsftpd.userlist  
userlist_deny=NO
```

```
user_config_dir=/etc/vsftpd_user_conf  
EOF
```

```
echo -e "alice\nmika" > /etc/vsftpd.userlist
```

```
mkdir -p /etc/vsftpd_user_conf  
echo "write_enable=NO" > /etc/vsftpd_user_conf/mika
```

```
service vsftpd restart
```

service vsftpd status

di alice

echo "nameserver 8.8.8.8" > /etc/resolv.conf

apt install -y --allow-unauthenticated ftp

cd /root

touch signal_alice.txt

ftp 10.190.2.2

```
root@alice:~# echo "nameserver 8.8.8.8" > /etc/resolv.conf
apt install -y --allow-unauthenticated ftp
cd /root
touch signal_alice.txt
ftp 10.190.2.2
ftp is already the newest version (20230507-2).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 18
Connected to 10.190.2.2.
220 (vsFTPD 3.0.5)
Name (10.190.2.2:root): alice
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> put signal_alice.txt
local: signal_alice.txt remote: signal_alice.txt
229 Entering Extended Passive Mode (|||38910|)
150 Ok to send data.
100% |*****| 13 87.55 KiB/s 00:00 ETA
226 Transfer complete.
13 bytes sent in 00:00 (6.71 KiB/s)
ftp> ls
229 Entering Extended Passive Mode (|||50354|)
150 Here comes the directory listing.
-rw-r--r-- 1 1000 1000 13 Sep 17 16:01 signal_alice.txt
226 Directory send OK.
ftp> quit
421 Timeout.
```

dimika

echo "nameserver 8.8.8.8" > /etc/resolv.conf

apt update --fix-missing

apt install -y --allow-unauthenticated ftp

cd /root

touch test_mika.txt

ftp 10.190.2.2

```

root@mika:~# echo "nameserver 8.8.8.8" > /etc/resolv.conf
apt update --fix-missing
apt install -y --allow-unauthenticated ftp
Get:1 http://deb.debian.org/debian trixie InRelease [140 kB]
Get:2 http://deb.debian.org/debian trixie-updates InRelease [47.3 kB]
Get:3 http://deb.debian.org/debian-security trixie-security InRelease [43.4 kB]
Get:4 http://deb.debian.org/debian trixie/main amd64 Packages [9678 kB]
Get:5 http://deb.debian.org/debian trixie-updates/main amd64 Packages [4412 B]
Get:6 http://deb.debian.org/debian-security trixie-security/main amd64 Packages [262 kB]
Fetched 10.2 MB in 1s (7158 kB/s)
19 packages can be upgraded. Run 'apt list --upgradable' to see them.
Installing:
  ftp

Installing dependencies:
  tftpd

Summary:
  Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 19
  Download size: 163 kB
  Space needed: 315 kB / 97.3 GB available

Get:1 http://deb.debian.org/debian trixie/main amd64 tftpd amd64 20230507-2+b1 [127 kB]
Get:2 http://deb.debian.org/debian trixie/main amd64 ftp all 20230507-2 [36.4 kB]
Fetched 163 kB in 0s (877 kB/s)
Selecting previously unselected package tftpd.
(Reading database ... 10187 files and directories currently installed.)
Preparing to unpack .../tftpd_20230507-2+b1_amd64.deb ...
Unpacking tftpd (20230507-2+b1) ...
Selecting previously unselected package ftp.
Preparing to unpack .../ftp_20230507-2_all.deb ...
Unpacking ftp (20230507-2) ...
Setting up tftpd (20230507-2+b1) ...
update-alternatives: using /usr/bin/tftpd to provide /usr/bin/ftp (ftp) in auto mode
update-alternatives: warning: skip creation of /usr/share/man/man1/ftp.1.gz because associated file /usr/share/man/man1/tftpd.1.gz (of link group ftp) doesn't exist
Setting up ftp (20230507-2) ...

```

```

root@mika:~# cd /root
touch test_mika.txt
ftp 10.190.2.2
Connected to 10.190.2.2.
220 (vsFTPd 3.0.5)
Name (10.190.2.2:root): mika
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> ls
229 Entering Extended Passive Mode (|||8014|)
150 Here comes the directory listing.
-rw-r--r--  1 1000    1000          13 Sep 17 16:01 signal_alice.txt
226 Directory send OK.
ftp> put test_mika.txt
local: test_mika.txt remote: test_mika.txt
229 Entering Extended Passive Mode (|||6601|)
550 Permission denied.
ftp> quit
221 Goodbye.

```

eiri

echo "nameserver 8.8.8.8" > /etc/resolv.conf

apt update --fix-missing

apt install -y --allow-unauthenticated ftp

ftp 10.190.2.2

```

root@eiri:~# echo "nameserver 8.8.8.8" > /etc/resolv.conf
apt update --fix-missing
apt install -y --allow-unauthenticated ftp
ftp 10.190.2.2
Get:1 http://deb.debian.org/debian trixie InRelease [140 kB]
Get:2 http://deb.debian.org/debian trixie-updates InRelease [47.3 kB]
Get:3 http://deb.debian.org/debian-security trixie-security InRelease [43.4 kB]
Get:4 http://deb.debian.org/debian trixie/main amd64 Packages [9678 kB]
Get:5 http://deb.debian.org/debian trixie-updates/main amd64 Packages [4412 B]
Get:6 http://deb.debian.org/debian-security trixie-security/main amd64 Packages [262 kB]
Fetched 10.2 MB in 1s (7468 kB/s)
19 packages can be upgraded. Run 'apt list --upgradable' to see them.
Installing:
  ftp

Installing dependencies:
  tftp

Summary:
  Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 19
  Download size: 163 kB
  Space needed: 315 kB / 97.3 GB available

Get:1 http://deb.debian.org/debian trixie/main amd64 tftp amd64 20230507-2+b1 [127 kB]
Get:2 http://deb.debian.org/debian trixie/main amd64 ftp all 20230507-2 [36.4 kB]
Fetched 163 kB in 0s (949 kB/s)
Selecting previously unselected package tftp.
(Reading database ... 10187 files and directories currently installed.)
Preparing to unpack .../tftp_20230507-2+b1_amd64.deb ...
Unpacking tftp (20230507-2+b1) ...
Selecting previously unselected package ftp.
Preparing to unpack .../ftp_20230507-2_all.deb ...
Unpacking ftp (20230507-2) ...
Setting up tftp (20230507-2+b1) ...
update-alternatives: using /usr/bin/tftp to provide /usr/bin/ftp (ftp) in auto mode
update-alternatives: warning: skip creation of /usr/share/man/man1/ftp.1.gz because associated file /usr/share/man/man1/tftp.1.gz (of link group ftp) doesn't exist
Setting up ftp (20230507-2) ...
Connected to 10.190.2.2.
220 (vsFTPD 3.0.5)
Name (10.190.2.2:root): eiri
pa530 Permission denied.
ftp: Login failed
ftp> password
?Invalid command.
ftp> quit
221 Goodbye.

```

8. Kelompok rahasia Knights perlu mengirimkan dokumen laporan intelijen ke FTP Server Chisa. Lakukan koneksi FTP client dari node Knights ke FTP Server Chisa menggunakan akun alice. Upload file berikut ([link file](#)). Analisis sesi Wireshark dan sebutkan: perintah FTP untuk upload (STOR), kode status sukses server (226), dan port data TCP yang dinegosiasikan pada mode PASV.

```

apt update --fix-missing
apt install -y --allow-unauthenticated ftp wget tshark
cd /root
wget --no-check-certificate
'https://docs.google.com/uc?export=download&id=1IFepK4wFmx55PnRki3NsHW-ivu
dSR0vg' -O knights_report.txt

```

```
tshark -i eth0 -f "host 10.190.2.2" -w /tmp/soal8.pcap >/dev/null 2>&1 &
```

```

Processing triggers for libc-bin (2.41-12+deb13u3) ...
root@knights:~# cd /root
root@knights:~# wget --no-check-certificate 'https://docs.google.com/uc?export=download&id=LINK_FILE_SOAL' -O knights_report.txt
--2026-09-16 02:21:10-- https://docs.google.com/uc?export=download&id=LINK_FILE_SOAL
Resolving docs.google.com (docs.google.com)... 74.125.200.100, 74.125.200.138, 74.125.200.139, ...
Connecting to docs.google.com (docs.google.com)[74.125.200.100]:443... connected.
HTTP request sent, awaiting response... 303 See other
Location: https://drive.usercontent.google.com/download?id=LINK_FILE_SOAL&export=download [following]
--2026-09-16 02:21:10-- https://drive.usercontent.google.com/download?id=LINK_FILE_SOAL&export=download
Resolving drive.usercontent.google.com (drive.usercontent.google.com)... 172.253.144.132, 2404:6800:4003:c24:84
Connecting to drive.usercontent.google.com (drive.usercontent.google.com)[172.253.144.132]:443... connected.
HTTP request sent, awaiting response... 404 Not Found
2026-09-16 02:21:11 ERROR 404: Not Found.

root@knights:~# wget --no-check-certificate 'https://docs.google.com/uc?export=download&id=https://drive.google.com/file/d/1lFepK4wFmx55PnRk13NsHW-ivudSR0vg/view?usp=sharing' -O
knights_report.txt
--2026-09-16 02:21:57-- https://docs.google.com/uc?export=download&id=https://drive.google.com/file/d/1lFepK4wFmx55PnRk13NsHW-ivudSR0vg/view?usp=sharing
Resolving docs.google.com (docs.google.com)... 74.125.200.101, 74.125.200.138, 74.125.200.139, ...
Connecting to docs.google.com (docs.google.com)[74.125.200.101]:443... connected.
HTTP request sent, awaiting response... 404 Not Found
2026-09-16 02:21:58 ERROR 404: Not Found.

root@knights:~# cd /root
wget --no-check-certificate 'https://docs.google.com/uc?export=download&id=1lFepK4wFmx55PnRk13NsHW-ivudSR0vg' -O knights_report.txt
--2026-09-16 02:22:29-- https://docs.google.com/uc?export=download&id=1lFepK4wFmx55PnRk13NsHW-ivudSR0vg
Resolving docs.google.com (docs.google.com)... 142.250.4.138, 142.250.4.139, 142.250.4.101, ...
Connecting to docs.google.com (docs.google.com)[142.250.4.138]:443... connected.
HTTP request sent, awaiting response... 303 See other
Location: https://drive.usercontent.google.com/download?id=1lFepK4wFmx55PnRk13NsHW-ivudSR0vg&export=download [following]
--2026-09-16 02:22:29-- https://drive.usercontent.google.com/download?id=1lFepK4wFmx55PnRk13NsHW-ivudSR0vg&export=download
Resolving drive.usercontent.google.com (drive.usercontent.google.com)... 172.253.144.132, 2404:6800:4003:c24:84
Connecting to drive.usercontent.google.com (drive.usercontent.google.com)[172.253.144.132]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 772 [application/octet-stream]
Saving to: 'knights_report.txt'

knights_report.txt      100%[=====] 772 --.-KB/s   in 0s

2026-09-16 02:22:31 (43.5 MB/s) - 'knights_report.txt' saved [772/772]

root@knights:~# tshark -i any -f "host 10.190.2.2" -w /root/soal8.pcap
Running as user "root" and group "root". This could be dangerous.
Capturing on "any"
tshark: Promiscuous mode not supported on the "any" device.
tshark: The file to which the capture would be saved ("/root/soal8.pcap") could not be opened: Permission denied.
root@knights:~# tshark -i eth0 -f "host 10.190.2.2" -w /tmp/soal8.pcap &
[1] 854
root@knights:~# Running as user "root" and group "root". This could be dangerous.
Capturing on "eth0"
ftp 10.190.2.2
Connected to 10.190.2.2.
220 (vsFTPd 3.0.5)
5 *[[C]][[C]][[C]][[C]][[C]][[C]]
8 ^C8knights:~#

```

```
knights - PuTTY
[2] 882
[1] Killed tshark -i eth0 -f "host 10.190.2.2" -w /tmp/soal8.pcap
root@knights:~# ftp 10.190.2.2
Connected to 10.190.2.2.
220 (vsFTPd 3.0.5)
Name (10.190.2.2:root): alice
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> pasv
7Invalid command.
ftp> passive
Passive mode: off; fallback to active mode: off.
ftp> put knights_report.txt
local: knights_report.txt remote: knights_report.txt
200 ERT command successful. Consider using EPSV.
150 Ok to send data.
100% |*****| 772 8.00 MiB/s 00:00 ETA
226 Transfer complete.
772 bytes sent in 00:00 (406.63 KiB/s)
ftp> quit
221 Goodbye.
root@knights:~# kill tshark
root@knights:~# tshark -r /tmp/soal8.pcap -Y "ftp || ftp-data"
Running as user "root" and group "root". This could be dangerous.
 4 0.004628335 10.190.2.2 ? 10.190.3.2 FTP 86 Response: 220 (vsFTPd 3.0.5)
 6 3.055873235 10.190.3.2 ? 10.190.2.2 FTP 78 Request: USER alice
 8 3.056758081 10.190.2.2 ? 10.190.3.2 FTP 100 Response: 331 Please specify the password.
10 5.506031273 10.190.3.2 ? 10.190.2.2 FTP 81 Request: PASS password
12 5.605019635 10.190.2.2 ? 10.190.3.2 FTP 89 Response: 230 Login successful.
14 5.605154674 10.190.3.2 ? 10.190.2.2 FTP 72 Request: SYST
16 5.605661487 10.190.2.2 ? 10.190.3.2 FTP 85 Response: 215 UNIX Type: L8
17 5.605850460 10.190.3.2 ? 10.190.2.2 FTP 72 Request: FEAT
18 5.606512996 10.190.2.2 ? 10.190.3.2 FTP 81 Response: 211-Features:
19 5.607220371 10.190.2.2 ? 10.190.3.2 FTP 73 Response: EPRT
20 5.607239968 10.190.2.2 ? 10.190.3.2 FTP 73 Response: EPSV
21 5.607247681 10.190.2.2 ? 10.190.3.2 FTP 73 Response: MDTM
22 5.607253895 10.190.2.2 ? 10.190.3.2 FTP 73 Response: PASV
23 5.607274218 10.190.2.2 ? 10.190.3.2 FTP 80 Response: REST STREAM
25 5.607280660 10.190.2.2 ? 10.190.3.2 FTP 73 Response: SIZE
26 5.607286918 10.190.2.2 ? 10.190.3.2 FTP 73 Response: TVFS
27 5.607293630 10.190.2.2 ? 10.190.3.2 FTP 75 Response: 211 End
29 47.456899991 10.190.3.2 ? 10.190.2.2 FTP 74 Request: TYPE I
30 47.457891463 10.190.2.2 ? 10.190.3.2 FTP 97 Response: 200 Switching to Binary mode.
31 47.458087899 10.190.3.2 ? 10.190.2.2 FTP 93 Request: ERT (|110.190.3.2|58983|)
32 47.458817695 10.190.2.2 ? 10.190.3.2 FTP 117 Response: 200 EPRT command successful. Consider using EPSV.
33 47.458919440 10.190.3.2 ? 10.190.2.2 FTP 91 Request: STOR knights_report.txt
37 47.461081756 10.190.2.2 ? 10.190.3.2 FTP 88 Response: 150 Ok to send data.
43 47.462891899 10.190.2.2 ? 10.190.3.2 FTP 90 Response: 226 Transfer complete.
45 67.457581029 10.190.3.2 ? 10.190.2.2 FTP 72 Request: QUIT
46 67.49655399 10.190.2.2 ? 10.190.3.2 FTP 80 Response: 221 Goodbye.
[2]+ Done tshark -i eth0 -f "host 10.190.2.2" -w /tmp/soal8.pcap > /dev/null 2>&1
root@knights:~#
```

Analisis & Jawaban Soal 8

- Perintah FTP untuk Upload: **STOR knights_report.txt** (Terlihat pada baris nomor 33: **Request: STOR knights_report.txt**).
- Kode Status Sukses Server: **226 Transfer complete** (Terlihat pada baris nomor 43: **Response: 226 Transfer complete**).
- Port Data TCP yang Dinegosiasikan pada Mode PASV: Saat mengetik perintah **passive**, koneksi berpindah ke mode Active (**EPRT**). Nah, untuk mendapatkan respon mode PASV, jalankan perintah berikut untuk melihat baris respon **227 Entering Passive Mode** beserta port-nya:
- Perintah FTP untuk Upload: **STOR knights_report.txt**
- Kode Status Sukses Server: **226 Transfer complete**
- Port Data TCP yang Dinegosiasikan: **59535** (Didapat dari respon **227 Entering Extended Passive Mode (|||59535|)**)

9. Mika mengakses dokumen Protokol Tujuh di ([link file](#)) dari FTP Server Chisa. Dari node Mika, unduh file tersebut menggunakan akun mika. Setelah itu, buktikan pembatasan read-only dengan mencoba mengunggah file baru dari akun mika, dan tunjukkan pesan error respon

server (error 550 Permission denied) saat mika mencoba melakukan upload.

```
cd /var/wired/data
```

```
wget --no-check-certificate
```

```
'https://docs.google.com/uc?export=download&id=1tKZu0rcti4t-fXX4jtXDSKDBWzsa  
wfoN' -O protocol7_manifesto.txt
```

```
chmod 644 /var/wired/data/protocol7_manifesto.txt
```

```
cd /root
```

```
touch test_upload_mika.txt
```

```
ftp 10.190.2.2
```

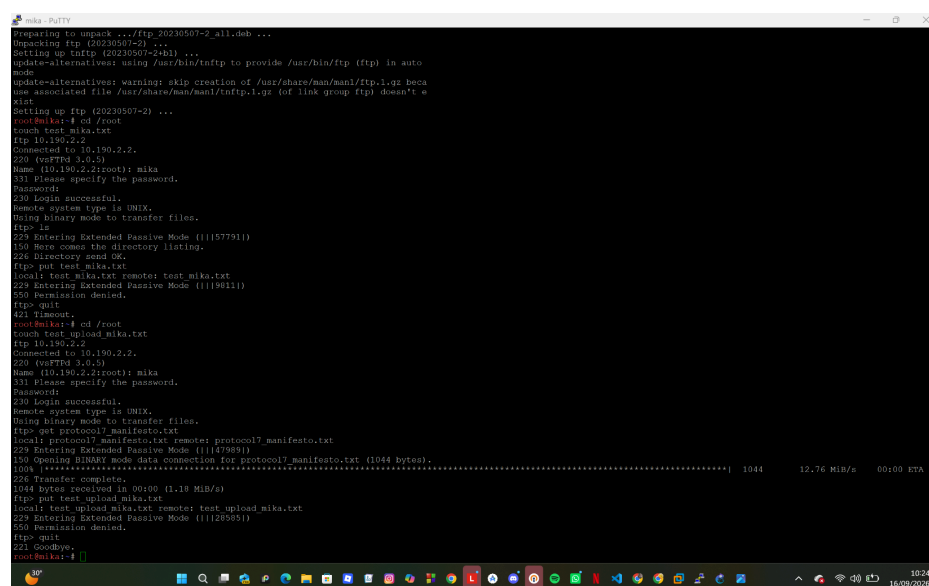
User: *mika* | Password: *password*

Perintah FTP:

```
get protocol7_manifesto.txt
```

```
put test_upload_mika.txt
```

```
quit
```



```
mika:PUTTY
Preparing to unpack .../ftp.20230507-2_all.deb ...
Unpacking ftp (20230507-2) ...
Setting up tnftp (20230507-2461) ...
update-alternatives: using /usr/bin/tnftp to provide /usr/bin/ftp (ftp) in auto
mode
update-alternatives: warning: skip creation of /usr/share/man/man1/ftp.1.gz beca
use associated file /usr/share/man/man1/tnftp.1.gz (of link group ftp) doesn't e
xist
Setting up ftp (20230507-2) ...
root@mika:~# cd /root
root@mika:~# touch test_mika.txt
root@mika:~# ftp 10.190.2.2
ftp 10.190.2.2:
Connected to 10.190.2.2.
220 (vsFTPD 3.0.5)
Name (10.190.2.2:root): mika
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> ls
229 Entering Extended Passive Mode (|||57791|)
150 Here comes the directory listing.
226 Directory send OK.
ftp> put test_mika.txt
local: test_mika.txt remote: test_mika.txt
229 Entering Extended Passive Mode (|||5811|)
550 Permission denied.
ftp> quit
421 Timeout.
root@mika:~# cd /root
root@mika:~# touch test_upload_mika.txt
root@mika:~# ftp 10.190.2.2
ftp 10.190.2.2:
Connected to 10.190.2.2.
220 (vsFTPD 3.0.5)
Name (10.190.2.2:root): mika
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> get protocol7_manifesto.txt
229 Entering Extended Passive Mode (|||47989|)
150 Opening BINARY mode data connection for protocol7_manifesto.txt (1044 bytes).
100% [*****] 1044 12.76 MiB/s 00:00 ETA
226 Transfer complete.
1044 bytes received in 00:00 (1.19 MiB/s)
ftp> put test_upload_mika.txt
local: test_upload_mika.txt remote: test_upload_mika.txt
229 Entering Extended Passive Mode (|||28585|)
550 Permission denied.
ftp> quit
221 Goodbye.
root@mika:~#
```

10. Knights melancarkan uji ketahanan koneksi ke server Chisa untuk menguji latensi jaringan The Wired. Kirimkan paket ping dari node Knights ke node Chisa dengan payload khusus 128 bytes dan interval 0.3 detik sebanyak 77 paket (ping -c 77 -s 128 -i 0.3 <IP_Chisa>). Buka Wireshark, catat nilai ICMP Type dan Code untuk Echo Request vs Echo Reply, serta analisis packet loss dan RTT (min/avg/max).

```
ping -c 77 -s 128 -i 0.3 10.190.2.2
```

1. Detail Header ICMP

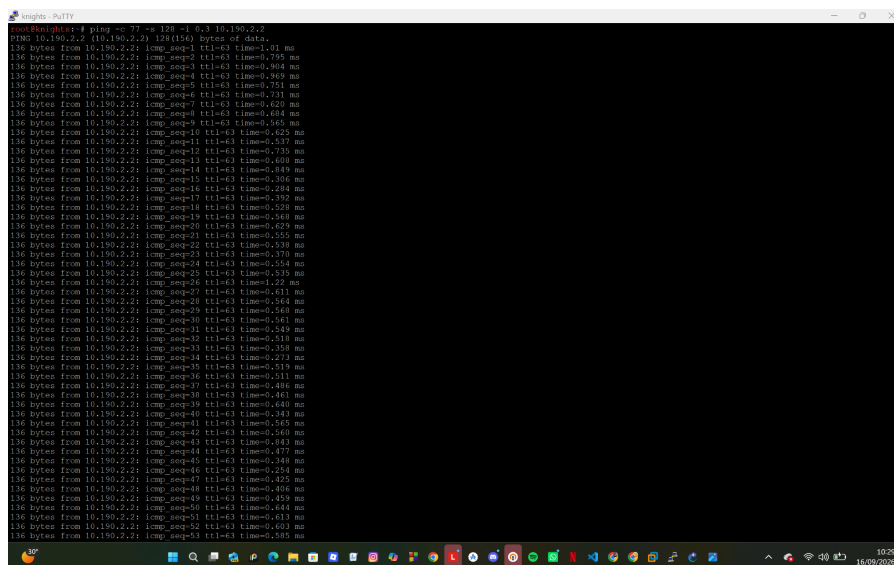
- Echo (Ping) Request (knights $\rightarrow$ chisa): Type 8, Code 0
- Echo (Ping) Reply (chisa $\rightarrow$ knights): Type 0, Code 0

2. Statistik Transmisi & Packet Loss

- Packets Transmitted: 77
- Packets Received: 77
- Packet Loss: 0%

3. Statistik Latensi Network (RTT / Round Trip Time)

- Min: 0.254 ms
- Avg: 0.586 ms
- Max: 1.216 ms
- mdev: 0.169 ms



```
knights: PuTTY
root@knights:~# ping -c 77 -s 128 -i 0.3 10.190.2.2
PING 10.190.2.2 (10.190.2.2) 128(1024) bytes of data:
 128 bytes from 10.190.2.2: icmp_seq=1 ttl=63 time=0.301 ms
 128 bytes from 10.190.2.2: icmp_seq=2 ttl=63 time=0.305 ms
 128 bytes from 10.190.2.2: icmp_seq=3 ttl=63 time=0.304 ms
 128 bytes from 10.190.2.2: icmp_seq=4 ttl=63 time=0.309 ms
 128 bytes from 10.190.2.2: icmp_seq=5 ttl=63 time=0.321 ms
 128 bytes from 10.190.2.2: icmp_seq=6 ttl=63 time=0.731 ms
 128 bytes from 10.190.2.2: icmp_seq=7 ttl=63 time=0.620 ms
 128 bytes from 10.190.2.2: icmp_seq=8 ttl=63 time=0.684 ms
 128 bytes from 10.190.2.2: icmp_seq=9 ttl=63 time=0.585 ms
 128 bytes from 10.190.2.2: icmp_seq=10 ttl=63 time=0.622 ms
 128 bytes from 10.190.2.2: icmp_seq=11 ttl=63 time=0.537 ms
 128 bytes from 10.190.2.2: icmp_seq=12 ttl=63 time=0.735 ms
 128 bytes from 10.190.2.2: icmp_seq=13 ttl=63 time=0.609 ms
 128 bytes from 10.190.2.2: icmp_seq=14 ttl=63 time=0.848 ms
 128 bytes from 10.190.2.2: icmp_seq=15 ttl=63 time=0.306 ms
 128 bytes from 10.190.2.2: icmp_seq=16 ttl=63 time=0.284 ms
 128 bytes from 10.190.2.2: icmp_seq=17 ttl=63 time=0.352 ms
 128 bytes from 10.190.2.2: icmp_seq=18 ttl=63 time=0.529 ms
 128 bytes from 10.190.2.2: icmp_seq=19 ttl=63 time=0.365 ms
 128 bytes from 10.190.2.2: icmp_seq=20 ttl=63 time=0.629 ms
 128 bytes from 10.190.2.2: icmp_seq=21 ttl=63 time=0.555 ms
 128 bytes from 10.190.2.2: icmp_seq=22 ttl=63 time=0.539 ms
 128 bytes from 10.190.2.2: icmp_seq=23 ttl=63 time=0.370 ms
 128 bytes from 10.190.2.2: icmp_seq=24 ttl=63 time=0.554 ms
 128 bytes from 10.190.2.2: icmp_seq=25 ttl=63 time=0.535 ms
 128 bytes from 10.190.2.2: icmp_seq=26 ttl=63 time=1.222 ms
 128 bytes from 10.190.2.2: icmp_seq=27 ttl=63 time=0.611 ms
 128 bytes from 10.190.2.2: icmp_seq=28 ttl=63 time=0.564 ms
 128 bytes from 10.190.2.2: icmp_seq=29 ttl=63 time=0.569 ms
 128 bytes from 10.190.2.2: icmp_seq=30 ttl=63 time=0.561 ms
 128 bytes from 10.190.2.2: icmp_seq=31 ttl=63 time=0.548 ms
 128 bytes from 10.190.2.2: icmp_seq=32 ttl=63 time=0.518 ms
 128 bytes from 10.190.2.2: icmp_seq=33 ttl=63 time=0.358 ms
 128 bytes from 10.190.2.2: icmp_seq=34 ttl=63 time=0.272 ms
 128 bytes from 10.190.2.2: icmp_seq=35 ttl=63 time=0.519 ms
 128 bytes from 10.190.2.2: icmp_seq=36 ttl=63 time=0.511 ms
 128 bytes from 10.190.2.2: icmp_seq=37 ttl=63 time=0.406 ms
 128 bytes from 10.190.2.2: icmp_seq=38 ttl=63 time=0.461 ms
 128 bytes from 10.190.2.2: icmp_seq=39 ttl=63 time=0.440 ms
 128 bytes from 10.190.2.2: icmp_seq=40 ttl=63 time=0.343 ms
 128 bytes from 10.190.2.2: icmp_seq=41 ttl=63 time=0.565 ms
 128 bytes from 10.190.2.2: icmp_seq=42 ttl=63 time=0.560 ms
 128 bytes from 10.190.2.2: icmp_seq=43 ttl=63 time=0.613 ms
 128 bytes from 10.190.2.2: icmp_seq=44 ttl=63 time=0.477 ms
 128 bytes from 10.190.2.2: icmp_seq=45 ttl=63 time=0.348 ms
 128 bytes from 10.190.2.2: icmp_seq=46 ttl=63 time=0.254 ms
 128 bytes from 10.190.2.2: icmp_seq=47 ttl=63 time=0.425 ms
 128 bytes from 10.190.2.2: icmp_seq=48 ttl=63 time=0.406 ms
 128 bytes from 10.190.2.2: icmp_seq=49 ttl=63 time=0.459 ms
 128 bytes from 10.190.2.2: icmp_seq=50 ttl=63 time=0.644 ms
 128 bytes from 10.190.2.2: icmp_seq=51 ttl=63 time=0.613 ms
 128 bytes from 10.190.2.2: icmp_seq=52 ttl=63 time=0.603 ms
 128 bytes from 10.190.2.2: icmp_seq=53 ttl=63 time=0.505 ms
 128 bytes from 10.190.2.2: icmp_seq=54 ttl=63 time=0.505 ms
```

11.

buka console chisa:

```
useradd -m -s /bin/ phantom_user
```

```
echo "phantom_user:wired_ghost" | chpasswd
```

```
apt update && apt install -y telnetd openbsd-inetd
```

```
nano /etc/inetd.conf (untuk hapus # dan jalanin/aktifin telnet)
```

```
service openbsd-inetd restart
```

lalu tekan kebel yg hubungin chisa dan eiri dan start capture

```
telnet 10.190.2.2
```

Masukkan username: phantom_user

Masukkan password: wired_ghost

Ketik exit setelah berhasil masuk.

PENJELASAN :

1. Pembuatan Akun & Konfigurasi Layanan di Chisa

Di node Chisa (10.190.2.2), pastikan paket telnetd dan openbsd-inetd sudah terinstal, lalu buat akun khusus untuk pengujian:

```
echo "nameserver 8.8.8.8" > /etc/resolv.conf
```

```
useradd -m -s /bin/ phantom_user
```

```
echo "phantom_user:wired_ghost" | chpasswd
```

```
apt update && apt install -y openbsd-inetd telnetd
```

```
nano /etc/inetd.conf (untuk hapus # dan jalanin/aktifin telnet)
```

```
service openbsd-inetd restart
```

lalu tekan kebel yg hubungin chisa dan eiri dan start capture

```
telnet 10.190.2.2
```

Masukkan username: *phantom_user*

Masukkan password: *wired_ghost*

Ketik *exit* setelah berhasil masuk

2. Pengujian Login dari Eiri ke Chisa

Buka terminal node Eiri, lalu menjalankan perintah Telnet menuju IP node Chisa:

telnet 10.190.2.2

- Masukkan username: *phantom_user*
- Masukkan password: *wired_ghost*
- Setelah berhasil masuk, ketik *exit* untuk mengakhiri sesi.

3. Analisis Kredensial Plain Text via Wireshark (Follow TCP Stream)

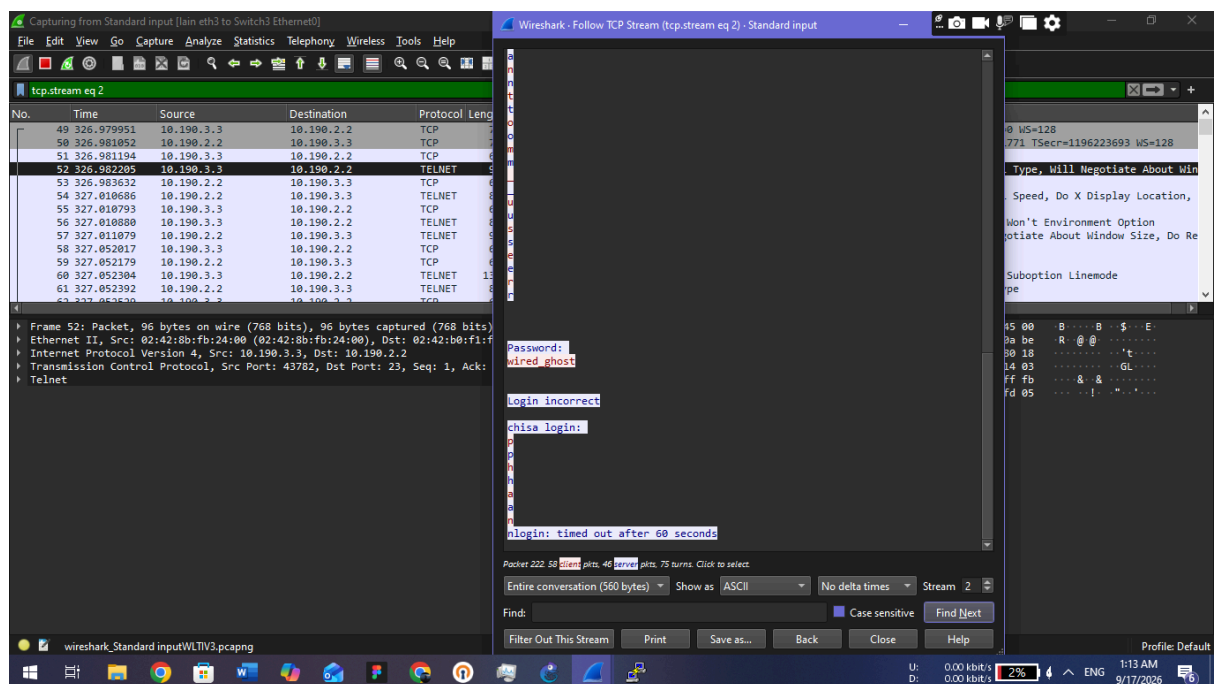
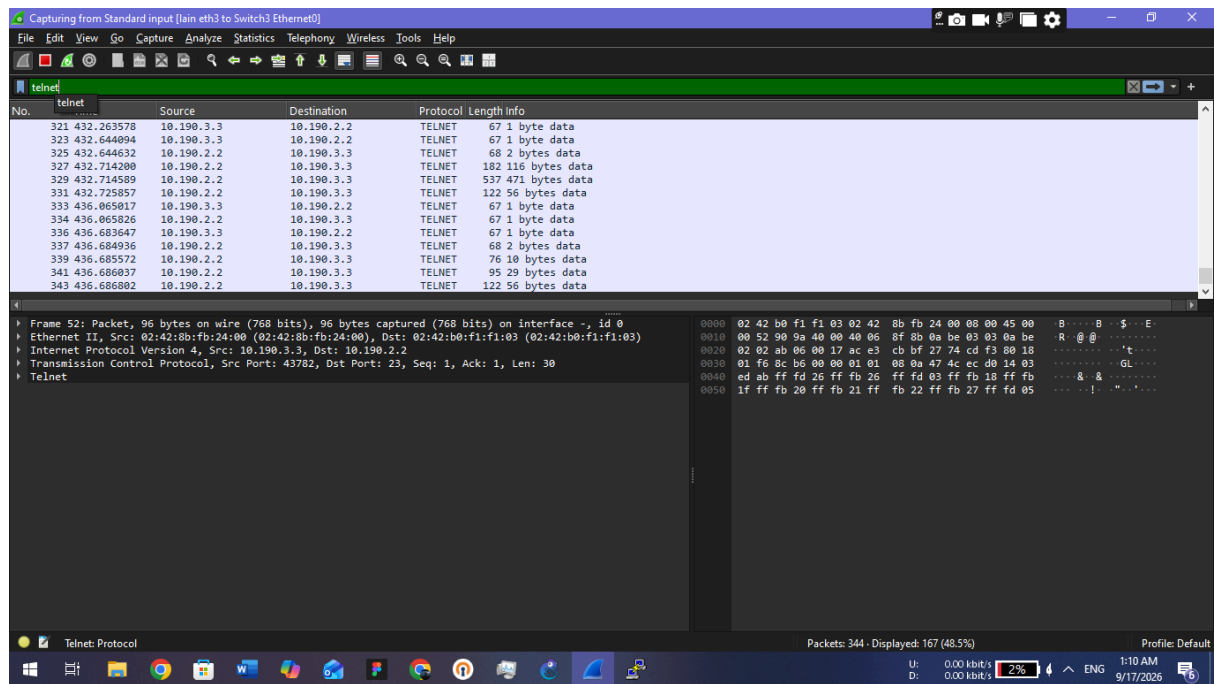
Saat sesi Telnet berlangsung dan ditangkap menggunakan Wireshark (atau packet capture pada link jaringan terkait) :

1. Klik kanan pada salah satu paket Telnet yang melintas.
2. Pilih Follow -> TCP Stream.
3. ada seluruh percakapan, termasuk username (*phantom_user*) dan password (*wired_ghost*), tercetak dengan jelas dalam bentuk teks biasa (plain text) tanpa enkripsi apa pun.

4. Mengapa Setiap Karakter Terkirim dalam Paket TCP Terpisah?

Protokol Telnet dirancang menggunakan mode interaktif per karakter (character-at-a-time atau NCP mode). Artinya, setiap kali menekan satu tombol di keyboard pada terminal klien (Eiri):

- Karakter tersebut langsung dibungkus menjadi sebuah segmen/paket TCP tersendiri dan dikirimkan secara seketika ke server (Chisa).
- Server kedian akan mengirimkan kembali paket acknowledgment sekaligus memantulkan (echo) karakter tersebut kembali ke layar klien agar bisa terlihat oleh pengguna. Inilah sebabnya dalam rekaman Wireshark, akan melihat banyak sekali paket TCP kecil berukuran beberapa byte yang masing-masing hanya berisi satu karakter ketikan (misalnya huruf 'p', lalu 'h', 'a', dan seterusnya secara berurutan).



12. 1. Atur Nameserver (DNS) di Knights: Langkah 1.

Di terminal node Knights, pake perintah berikut untuk menyetel DNS Google:

```
echo "nameserver 8.8.8.8" > /etc/resolv.conf
```

2. Install SSH & Apache2: Langkah 2.

Setelah DNS diatur, kita jalankan ulang perintah update dan instalasinya:

apt update && apt install -y openssh-server apache2

3.Jalankan Layanan:Langkah 3.

Nyalakan layanannya agar port 22 dan 80 aktif:

service ssh start

service apache2 start

Buka Terminal Alice & Jalankan Port Scanning:Langkah 1.

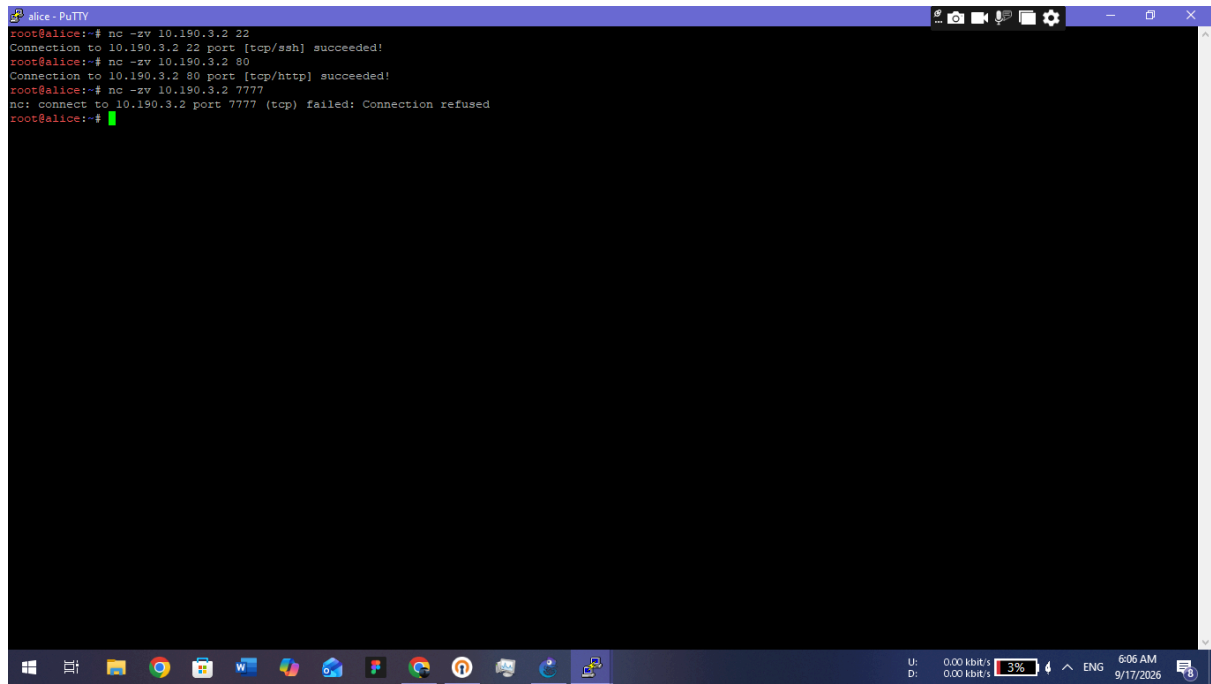
*Buka terminal node Alice di GNS3, lalu jalankan perintah Netcat (**nc**) dengan opsi **-zv** (z = scanning mode tanpa mengirim data, v = verbose) untuk memeriksa port 22, 80, dan 7777 pada node Knights (**10.190.2.2**):*

nc -zv 10.190.3.2 22

nc -zv 10.190.3.2 80

nc -zv 10.190.3.2 7777

Hasil di Terminal:



```
Alice - PuTTY
root@alice:~# nc -zv 10.190.3.2 22
Connection to 10.190.3.2 22 port [tcp/ssh] succeeded!
root@alice:~# nc -zv 10.190.3.2 80
Connection to 10.190.3.2 80 port [tcp/http] succeeded!
root@alice:~# nc -zv 10.190.3.2 7777
nc: connect to 10.190.3.2 port 7777 (tcp) failed: Connection refused
root@alice:~#
```

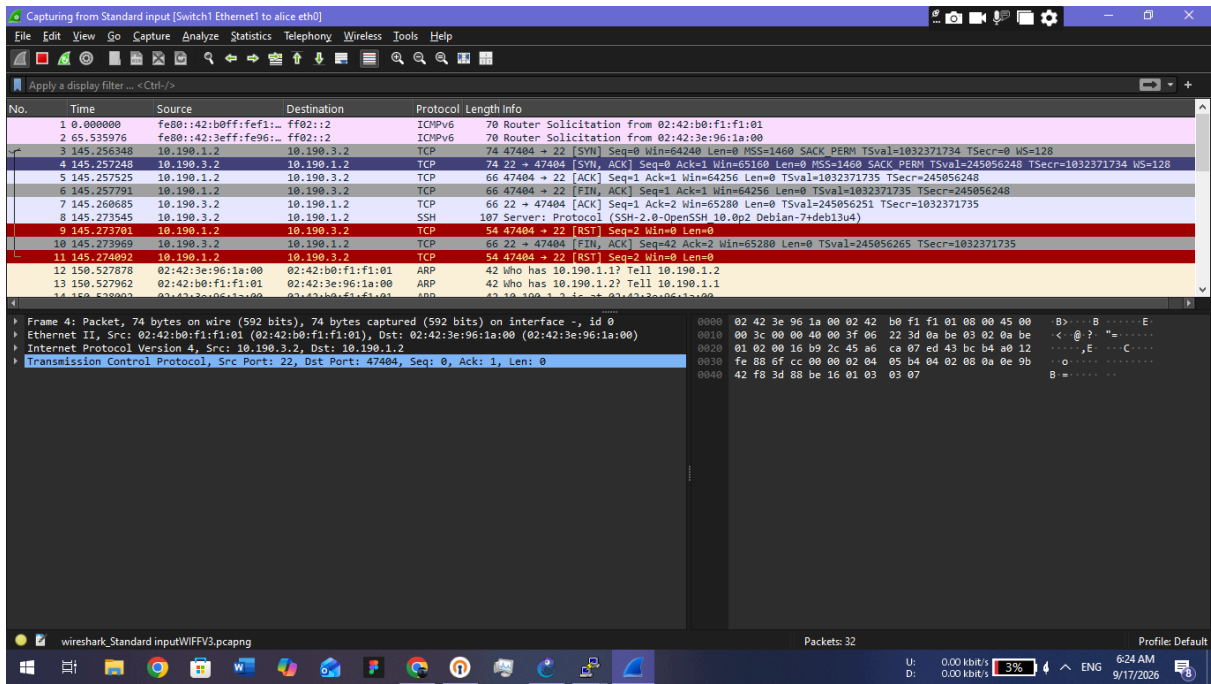
2. Analisis Perbedaan TCP Flag di Wireshark: Langkah 2.

Untuk menangkap paket pada skenario pemindaian port dari Alice ke Knights ini, kabel yang harus klik di GNS3 adalah:

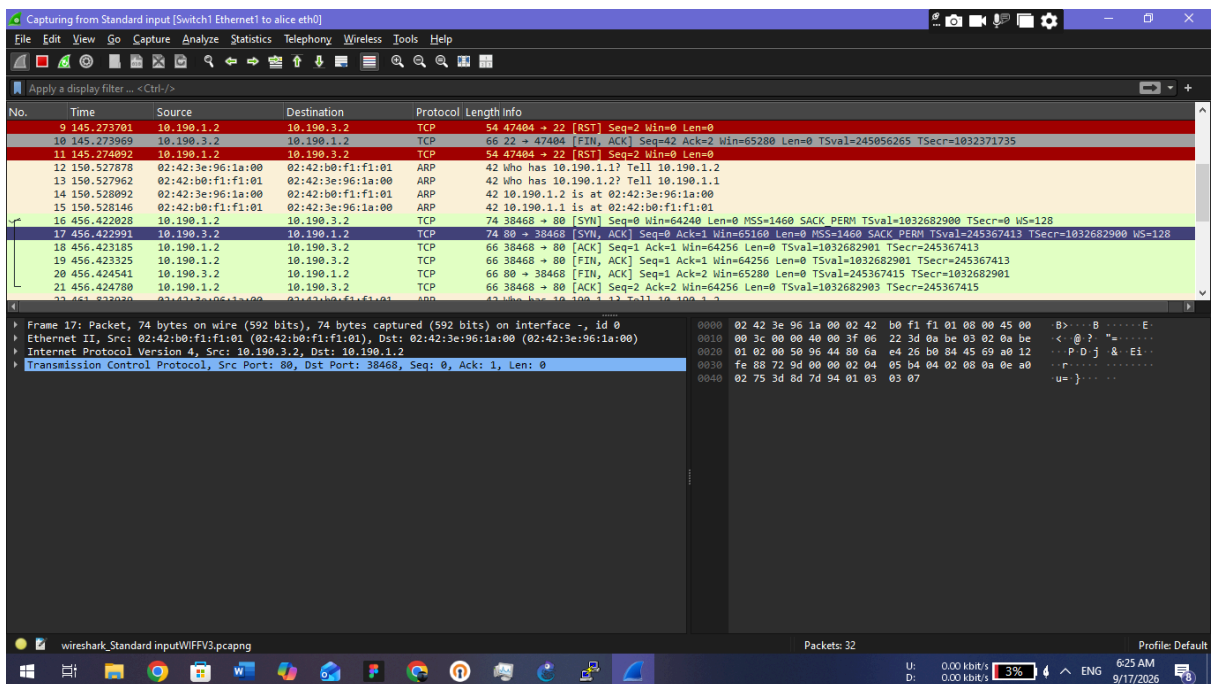
- Kabel antara Switch1 dan Alice (yang menghubungkan Switch1 port e1 ke *alice eth0*)
 - Atau
- Kabel antara Switch3 dan Knights (yang menghubungkan Switch3 port e1 ke *knights eth0*)

Jika merekam lalu lintas jaringan ini menggunakan Wireshark (atau *tshark*), perhatikan respons paket dari target (10.190.3.2) terhadap paket *SYN* yang dikirimkan oleh Alice:

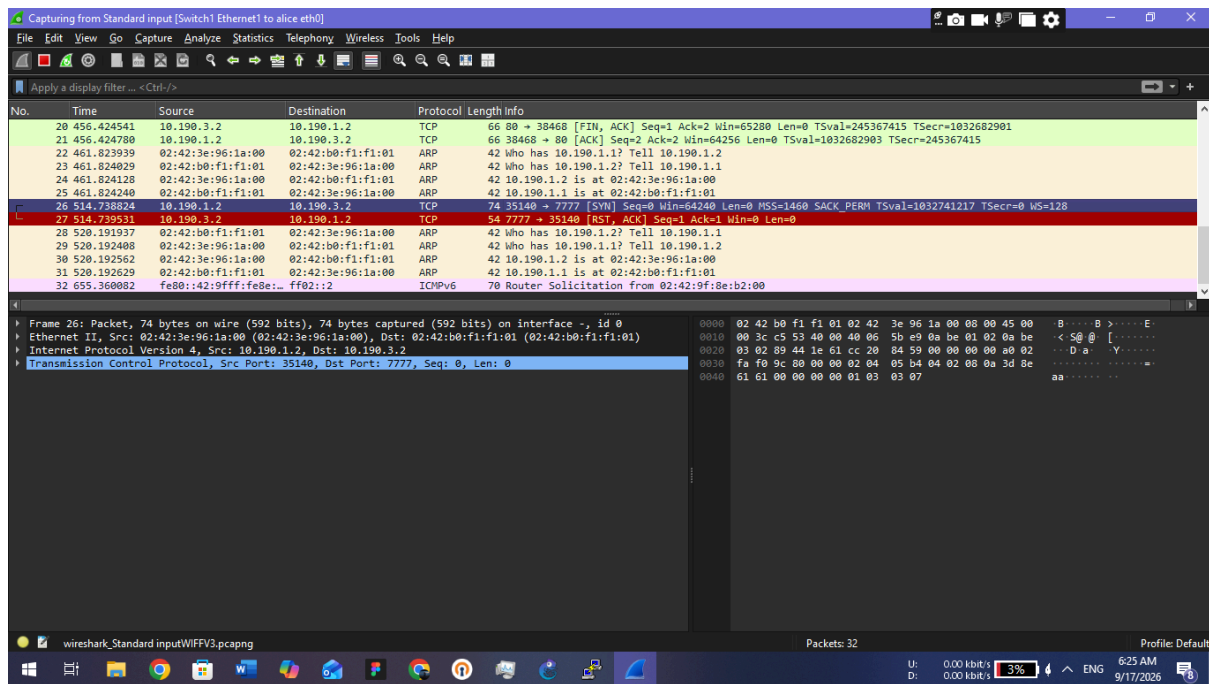
```
nc -zv 10.190.3.2 22
```



`nc -zv 10.190.3.2 80`



`nc -zv 10.190.3.2 7777`



- Untuk Port Terbuka (Port 22 & 80):
 - Server Knights akan merespons paket **SYN** dari Alice dengan mengirimkan kombinasi flag SYN-ACK (Synchronization-Acknowledgment). Ini menkan port tersebut aktif dan siap menerima koneksi (melanjutkan proses Three-Way Handshake).
- For Port Tertutup (Port 7777):
 - Server Knights akan langsung menolak koneksi dengan mengirimkan paket berflag RST-ACK (Reset-Acknowledgment). Ini adalah sinyal bahwa port tersebut sedang tidak mendengarkan (not listening) atau tertutup.

13. Langkah 1: Konfigurasi di Node Knights (Server SSH)

Buka terminal node Knights (10.190.3.2) untuk memastikan layanan SSH terinstal, lalu atur konfigurasinya agar hanya mengizinkan Public Key Authentication dan menolak login dengan password.

1.Install OpenSSH Server:Langkah

`apt update && apt install -y openssh-server`

2.Buat User mika_admin (Jika Belum Ada):Langkah

```
useradd -m -s /bin/ mika_admin
```

```
passwd mika_admin
```

```
password : mika_admin
```

(Masukkan password yang diinginkan untuk inisialisasi awal user).

3.Edit Konfigurasi SSHD:Langkah 1.3.

Buka file konfigurasi SSH:

```
nano /etc/ssh/sshd_config
```

Cari dan pastikan baris-baris berikut diubah/diaktifkan (hapus t pagar # di depannya jika ada):

```
PubkeyAuthentication yes
```

```
PasswordAuthentication no
```

```
PermitRootLogin no
```

(Simpan file dengan menekan **Ctrl+O**, **Enter**, lalu keluar dengan **Ctrl+X**).

4.Restart Layanan SSH:Langkah 1.4.

```
service ssh restart
```

Langkah 2: Buat SSH Key Pair di Node Mika (Client)

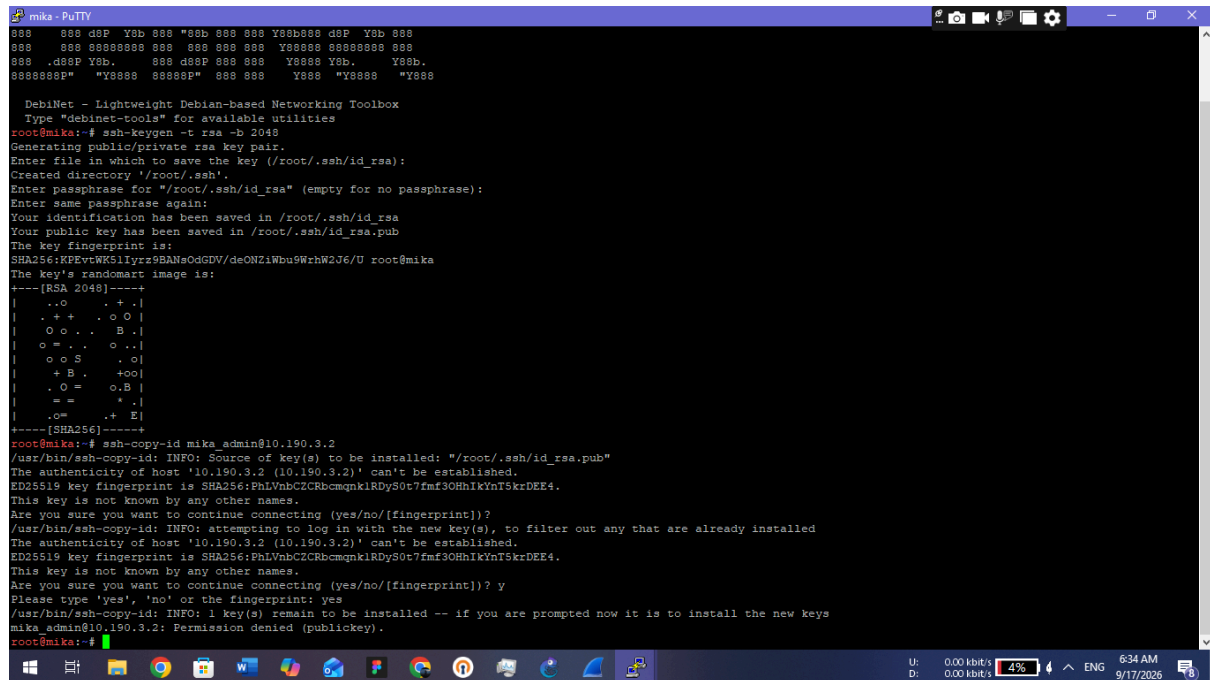
Buka terminal node Mika (masuk sebagai user **mika_admin** atau buat user tersebut terlebih dahulu di Mika jika diperlukan).

1.Generate SSH Key:Langkah 2.1.

Jalankan perintah berikut untuk membuat pasangan kunci (public dan private key):

`ssh-keygen -t rsa -b 2048`

(Tekan **Enter** terus menerus saat diminta file path dan passphrase hingga key berhasil digenerate).



```
mika - PuTTY
888 888 d8P Y8b 888 "88b 888 888 Y8b888 d8P Y8b 888
888 888 8888888 888 888 888 888 Y8888 8888888 888
888 .d88P Y8b. 888 d88P 888 888 Y8888 Y8b. Y88b.
8888888P" "Y888 88888P" 888 888 Y888 "Y8888 "Y888

DebiNet - Lightweight Debian-based Networking Toolbox
Type "debinet-tools" for available utilities
root@mika:~# ssh-keygen -t rsa -b 2048
Generating public/private rsa key pair.
Enter file in which to save the key (/root/.ssh/id_rsa):
Created directory '/root/.ssh'.
Enter passphrase for "/root/.ssh/id_rsa" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /root/.ssh/id_rsa
Your public key has been saved in /root/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:KPEvtWKS1lyrz9EANAoOdGV/deONZiWbuSWhW2J6/U root@mika
The key's randomart image is:
----[RSA 2048]-----
| ..o . + + |
| . + + . o o |
| o o . . B . |
| o = . . o . |
| o o S . o |
| + B . +oo |
| . O = o.B |
| = = * . |
| .Om . + E |
-----[SHA256]-----
root@mika:~# ssh-copy-id mika_admin@10.190.3.2
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/root/.ssh/id_rsa.pub"
The authenticity of host '10.190.3.2 (10.190.3.2)' can't be established.
ED25519 key fingerprint is SHA256:PhLVnbCZCRbcmqkLRDyS0t7fmf3OHhIkYnT5krDEE4.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])?
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
The authenticity of host '10.190.3.2 (10.190.3.2)' can't be established.
ED25519 key fingerprint is SHA256:PhLVnbCZCRbcmqkLRDyS0t7fmf3OHhIkYnT5krDEE4.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? y
Please type 'yes', 'no' or the fingerprint: yes
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
mika_admin@10.190.3.2: Permission denied (publickey).
root@mika:~#
```

2.Kirim Public Key ke Node Knights:Langkah 2.2.

Salin public key dari Mika ke node Knights (**10.190.3.2**) menggunakan perintah **ssh-copy-id**:

`ssh-copy-id mika_admin@10.190.3.2`

(Masukkan password user **mika_admin** yang tadi dibuat di Knights untuk otorisasi pengiriman kunci pertama kali).

1.Cek & Buat Direktori .ssh di Knights:Langkah 1.

Masuk ke terminal node Knights, lalu pastikan folder **.ssh** dan file **authorized_keys** miliki user **mika_admin** sudah ada dengan izin (permission) yang benar:

`su - mika_admin`

```
mkdir -p ~/.ssh
```

```
chmod 700 ~/.ssh
```

```
touch ~/.ssh/authorized_keys
```

```
chmod 600 ~/.ssh/authorized_keys
```

```
exit
```

2.Periksa Konfigurasi sshd_config di Knights:Langkah 2.

Buka file konfigurasi SSH:

```
nano /etc/ssh/sshd_config
```

Pastikan baris-baris berikut aktif (tidak ada t # di depannya) dan bernilai benar:

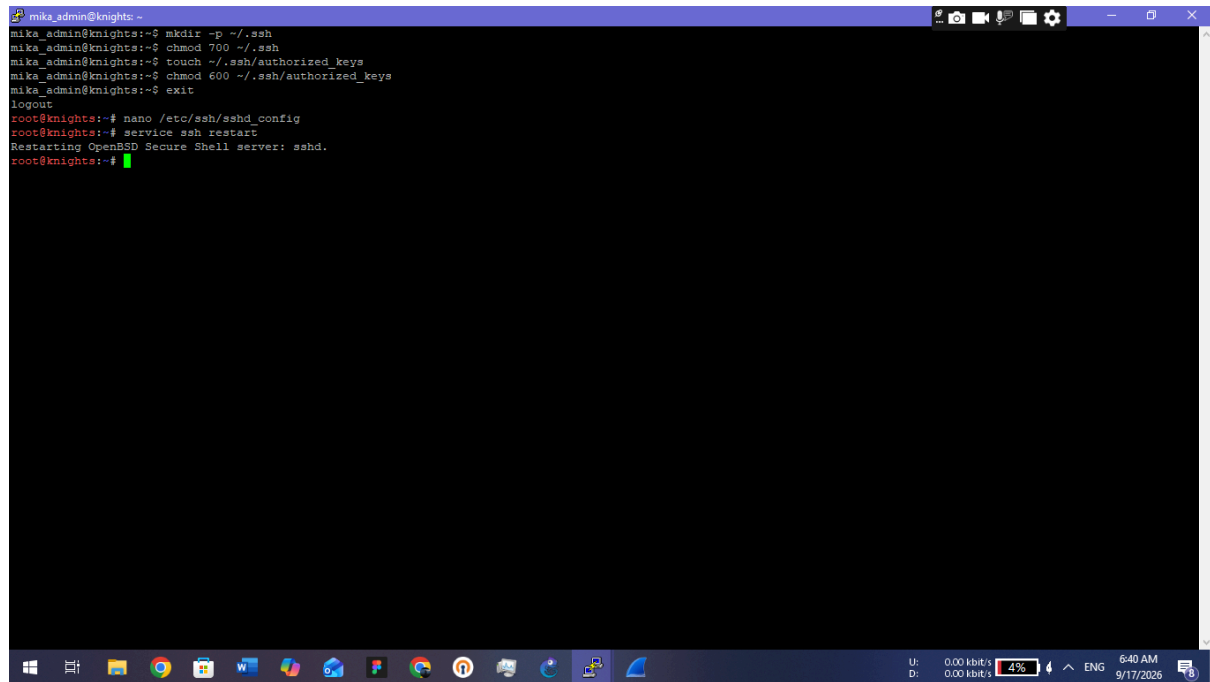
```
PubkeyAuthentication yes
```

```
AuthorizedKeysFile .ssh/authorized_keys
```

*(Simpan dengan **Ctrl+O**, **Enter**, lalu keluar dengan **Ctrl+X**).*

3.Restart Layanan SSH:Langkah 3.

```
service ssh restart
```



```
mika_admin@knights:~$ mkdir -p ~/.ssh
mika_admin@knights:~$ chmod 700 ~/.ssh
mika_admin@knights:~$ touch ~/.ssh/authorized_keys
mika_admin@knights:~$ chmod 600 ~/.ssh/authorized_keys
mika_admin@knights:~$ exit
logout
root@knights:~# nano /etc/ssh/sshd_config
root@knights:~# service ssh restart
Restarting OpenBSD Secure Shell server: sshd.
root@knights:~#
```

1. Tampilkan Public Key di Mika:Langkah 1.

Di terminal node Mika, jalankan perintah ini untuk melihat dan menyalin isi kunci publik:

```
cat /root/.ssh/id_rsa.pub
```

(Blok dan salin / copy teks panjang yang ncul, lai dari **ssh-rsa ...** sampai selesai).

2. Tempelkan ke Authorized_Keys di Knights:Langkah 2.

Buka terminal node Knights, lalu masukkan perintah

```
mkdir -p /home/mika_admin/.ssh
```

```
nano /home/mika_admin/.ssh/authorized_keys
```

(Tempel / paste teks kunci publik yang sudah salin dari Mika ke dalam file tersebut, simpan dengan menekan **Ctrl+O**, **Enter**, lalu keluar dengan **Ctrl+X**).

3. Atur Hak Akses di Knights:Langkah 3.

Pastikan kepemilikan dan perizinan foldernya aman dengan perintah:

```
chown -R mika_admin:mika_admin /home/mika_admin/.ssh
```

```
chmod 700 /home/mika_admin/.ssh
```

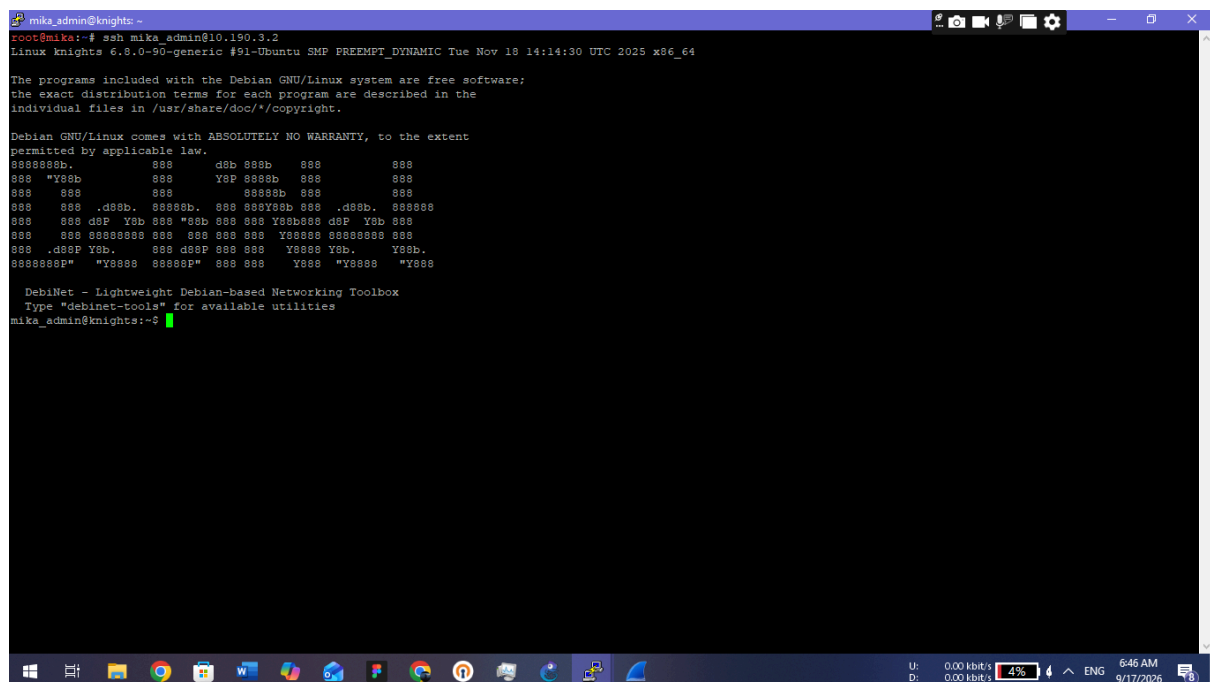
```
chmod 600 /home/mika_admin/.ssh/authorized_keys
```

Langkah 3: Uji Coba Koneksi SSH Tanpa Password

Dari terminal node Mika, coba lakukan koneksi SSH ke node Knights:

```
ssh mika_admin@10.190.3.2
```

- Hasil yang diharapkan: Mika akan langsung masuk ke terminal Knights tanpa diminta memasukkan password sama sekali berkat autentikasi kunci publik (public key). Ketik **exit** untuk kembali ke Mika.

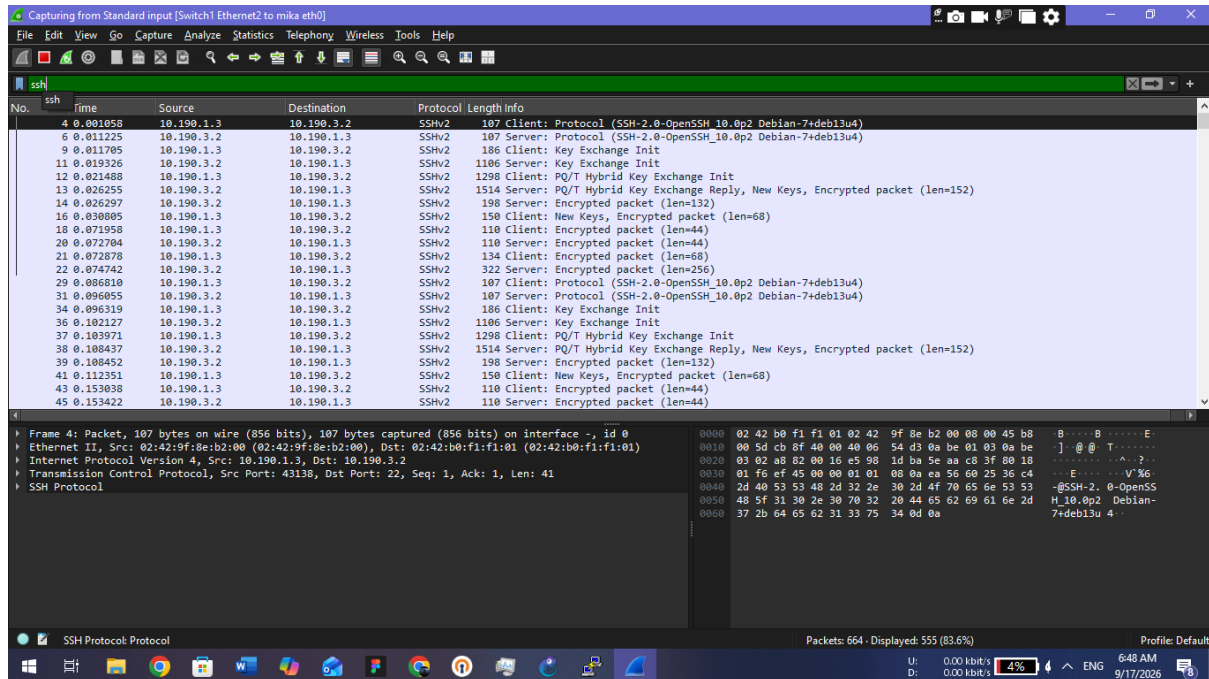


Langkah 4: Analisis Wireshark & Penjelasan Laporan

Untuk menjawab instruksi analisis di Wireshark, lakukan langkah-langkah berikut:

1. *Iai Capture Wireshark: Pasang tangkapan paket di kabel yang menghubungkan Mika atau Knights.*
2. *Jalankan Koneksi: Lakukan perintah **ssh mika_admin@10.190.3.2** dari Mika.*
3. *Cari Elemen di Wireshark (Gunakan filter **ssh**):*

- **Protocol Version Exchange:** Terlihat di awal koneksi berupa teks plain text sederhana yang saling bertukar informasi versi software (contoh: **SSH-2.0-OpenSSH_...**). Ini wajar karena kedua belah pihak harus tahu versi masing-masing sebelum enkripsi aktif.
- **Key Exchange (KEX):** Paket selanjutnya memperlihatkan proses negosiasi algoritma kriptografi, pertukaran kunci Diffie-Hellman (**Client Key Exchange, Server Key Exchange**), hingga ncul paket terenkripsi (**Encrypted Packet**).



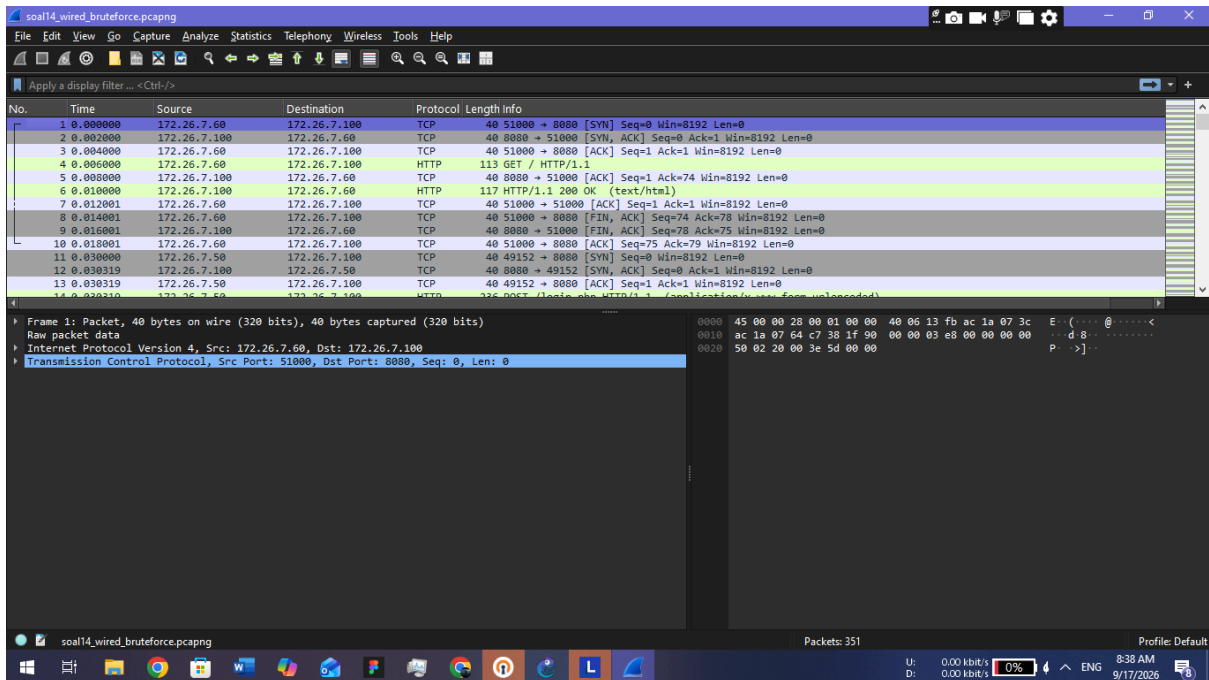
Jawaban Mengapa Kredensial Tidak Terlihat Seperti pada Telnet:

- **Telnet:** Mengirimkan seluruh data (termasuk username dan password) secara plain text (mentah) di atas TCP tanpa pengamanan apa pun, sehingga siapa pun yang menyadap (sniffing) di jalur jaringan bisa membaca teksnya secara langsung.
- **SSH:** Setelah tahap Key Exchange dan verifikasi Public Key selesai, seluruh sesi koneksi langsung dienkripsi secara end-to-end menggunakan algoritma simetris (seperti AES). Oleh karena itu, data sensitif seperti password (jika ada) atau perintah/karakter ketikan di dalam sesi SSH berubah menjadi ciphertext acak yang tidak bisa dibaca oleh penyadap jaringan.

14. Langkah 1: Buka File **.pcapng** di Wireshark

1. Unduh file **soal14_wired_bruteforce.pcapng** dari link Google Drive yang diberikan.

2. Buka aplikasi Wireshark, lalu Drag and Drop file tersebut ke Wireshark (atau pilih File > Open).

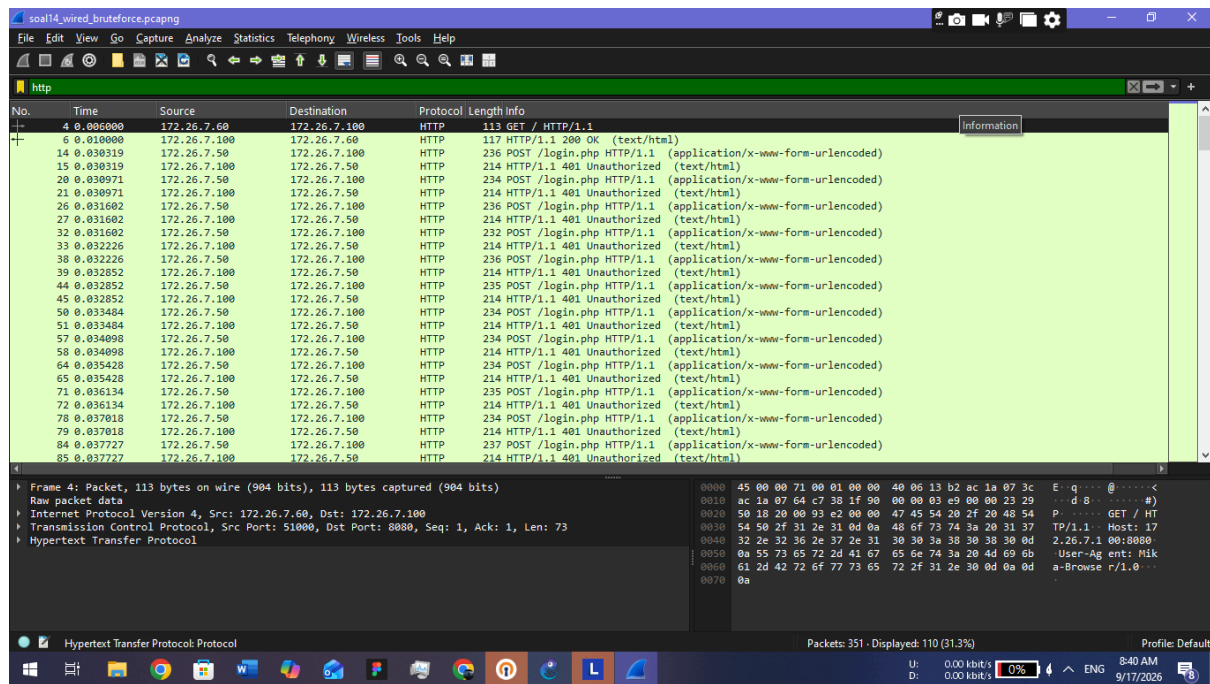


Langkah 2: Analisis Tean di Wireshark

Gunakan filter pencarian di bagian atas Wireshark untuk menekan parameter berikut:

1. Alamat IP Penyerang & Target (Beserta Port):

- Ketik filter di Wireshark: `http` atau `tcp.port == 80` (atau port web server yang digunakan).
- IP Penyerang: Lihat alamat IP dari node Eiri yang mengirimkan banyak permintaan HTTP secara beruntun (biasanya terlihat pengulangan POST request ke form login).
- Target IP & Port: Lihat alamat IP Alice (sebagai web server) dan port yang diserang (biasanya port `80` untuk HTTP atau `443` untuk HTTPS).



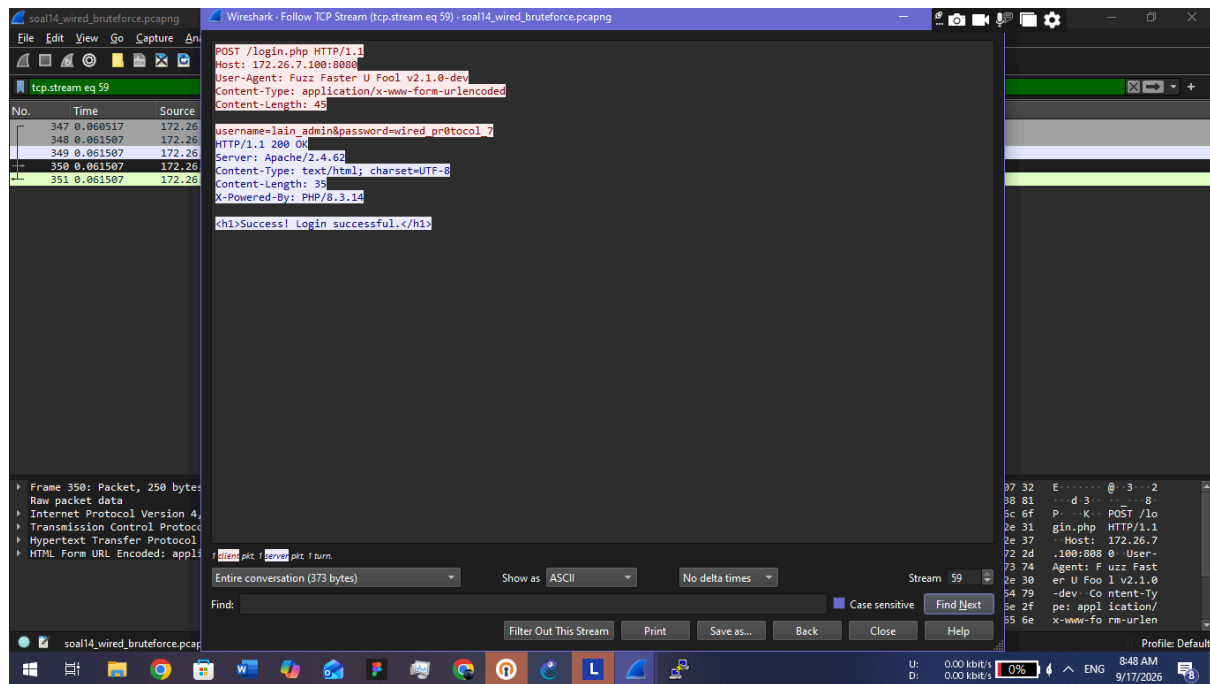
IP Penyerang (Attacker): 172.26.7.50 (karena IP ini yang terus-menerus mengirimkan POST /login.php secara beruntun).

Target IP: 172.26.7.100 (IP server yang diserang).

Port yang Diserang: Berdasarkan panel bawah bagian Transmission Control Protocol, target menggunakan Port 8080 (**Dst Port: 8080**).

2. Password **lain_admin** yang Berhasil Ditembus:

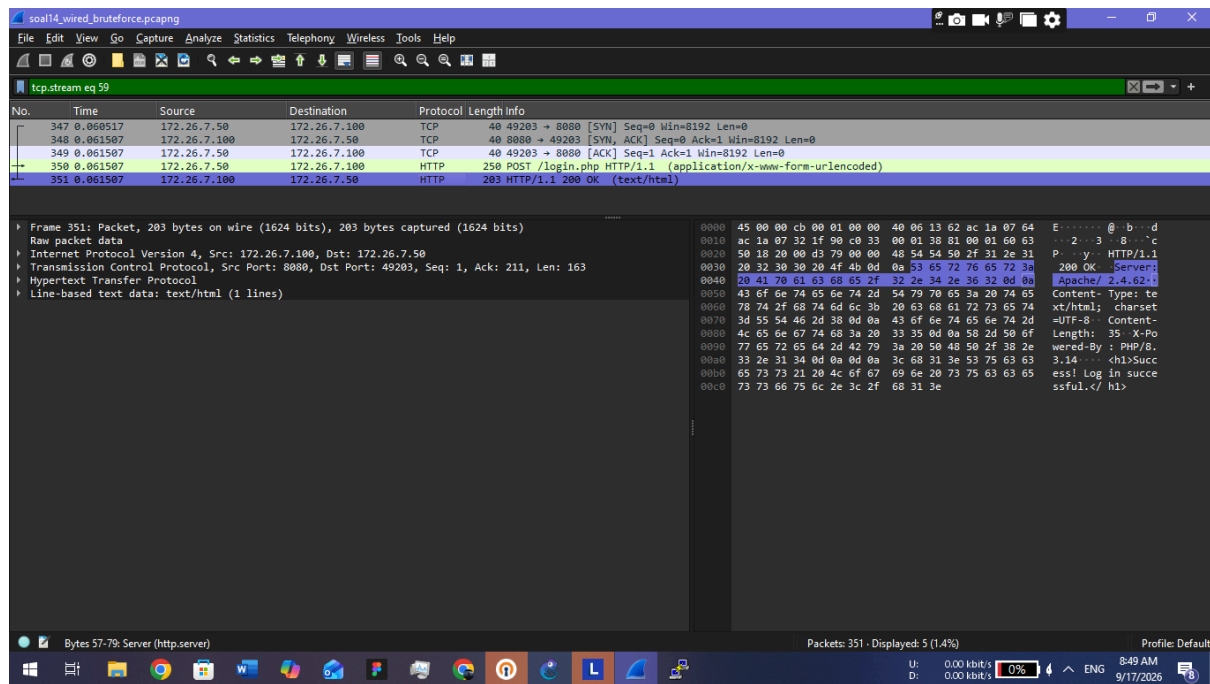
- Ketik filter untuk melihat isi paket HTTP POST:
http.request.method == "POST" atau klik kanan pada salah satu paket HTTP POST > Follow > TCP Stream.
- Gulir (scroll) ke bawah pada aliran stream tersebut untuk melihat percobaan brute-force beruntun hingga ditekan kombinasi password untuk user **lain_admin** yang memberikan respons sukses (misalnya kode status HTTP **200 OK** atau redirect, berbeda dari percobaan sebelumnya yang gagal). Password yang berhasil tembus biasanya ada di baris percobaan terakhir user tersebut.



Password = **wired_pr0tocol_7**

3. Web Server Software & Versi pada Response Header:

- Cari paket balasan HTTP dari server (biasanya bertuliskan **HTTP/1.1 200 OK** atau **302 Found**).
- Klik paket tersebut, lalu lihat bagian Hypertext Transfer Protocol di panel tengah.
- Cari baris **Server:** (contoh: **Apache/2.4.38 (Debian)**, **nginx/1.14.2**, dll.) untuk mencatat software web server beserta versi persisnya.



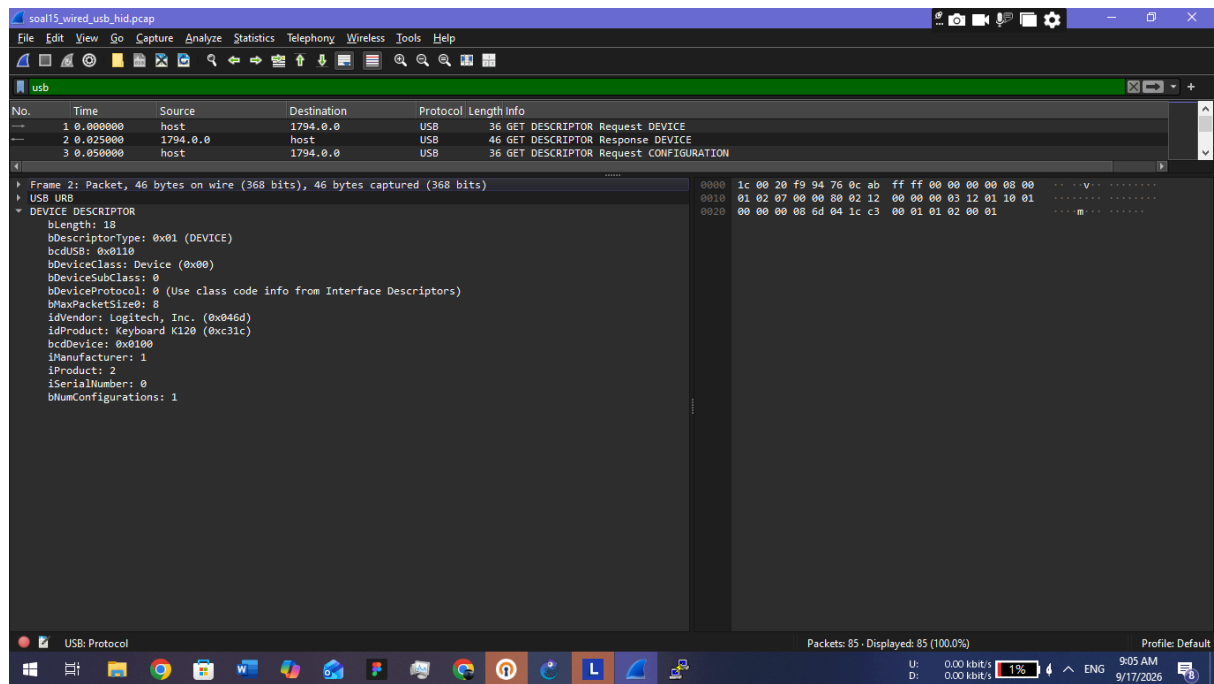
Server = Apache/2.4.62

Langkah 3: Kirim Jawaban ke Socket Server

Setelah sea data terkumpul (IP Penyerang, IP Target & Port, Password lain_admin, serta Software Web Server beserta versinya), buka terminal dan hubungkan ke socket server yang disediakan:

`nc [IP_Group] 3401`

Lalu masukkan jawaban sesuai format atau pertanyaan yang diminta oleh server interaktif tersebut.

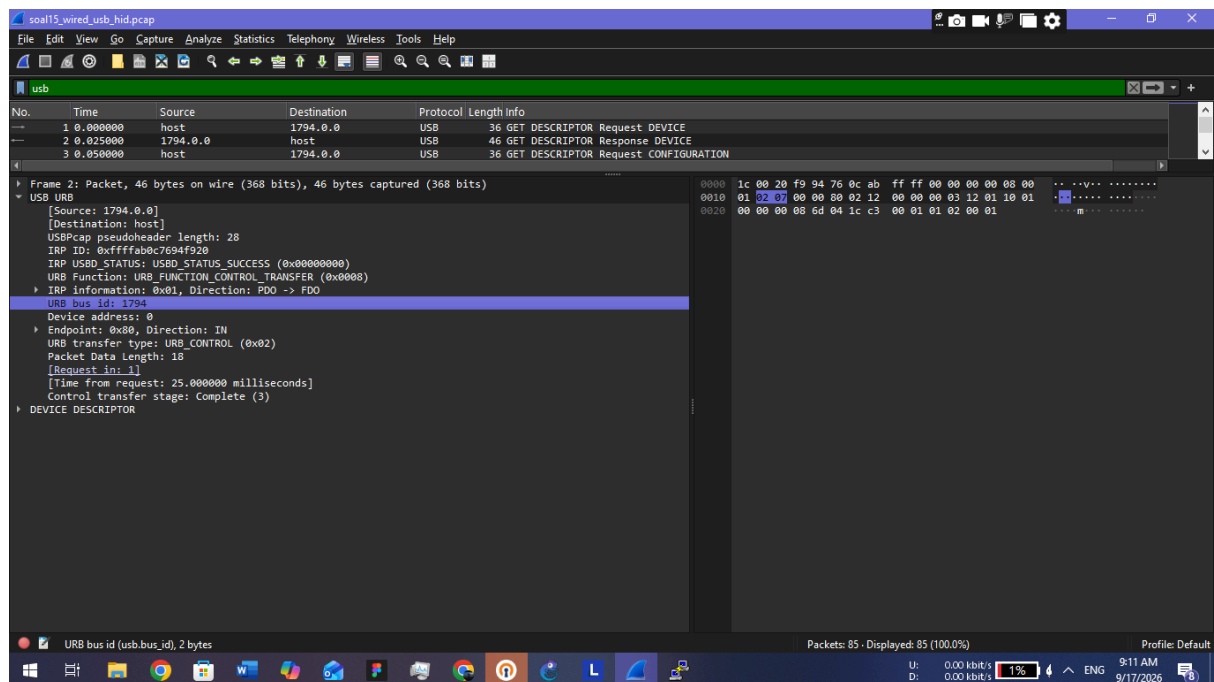


idVendor = Logitech, Inc. (0x046d)

idProduct = Keyboard K120 (0xc31c)

2. Alamat Nomor Device USB (Device Address):

- Di dalam paket penjelas perangkat USB yang sama (atau paket USB URB awal), cari atribut bernama **Bus ID** dan **Device Address** (biasanya berupa angka desimal kecil seperti 3 atau 2). Nilai Device Address inilah yang ditanyakan.



```
scancode_map = {
```

}

```
def parse_text_file(filename):
```

```
message = ""
```

with open(filename, 'r', encoding='utf-8', errors='ignore') as f:

```
for line in f:
```

```
# Cari baris yang mengandung Leftover Capture Data
```

if "Leftover Capture Data:" in line or "capdata:" in line:

```
parts = line.split(":")
```

```
if len(parts) > 1:
```

```
hex_str = parts[1].strip().replace(":", "").replace(" ", "")
```

```
if len(hex_str) >= 6:
```

Ambil byte ke-3 (karakter ke 4 dan 5 di string hex)

try:

```
scancode hex = hex str[4:6]
```

```
scancode = int(scancode_hex, 16)
```

```
if scancode in scancode_map:
```

```

        message += scancode_map[scancode]

    except ValueError:

        continue

print("=== PESAN RAHASIA USB HID ===")

print(message)

parse_text_file("data_usb.txt")

```

Cara Pakai:

1. *Simpan skrip di atas, lalu jalankan lewat terminal/Command Prompt:*
- 2.

```
python decode_usb.py
```

- 3.
- 4.
5. *Skrip akan otomatis membaca file **.pcap**, memfilter data USB, menerjemahkan kodenya, dan menampilkan teks pesan rahasianya secara utuh di layar terminal!*

CODE = wiredprotocol7isalive2026

flag: KOMJAR26{USB_K3ystr0k3_zWwnYRYEEQXKwDvQBtrYQ0mHf}

```
alice-PUTTY
root@alice:~# nc 10.4.89.250 3402

===== Soal 15 - Protocol 7: USB Keystroke Decoding =====
Difficulty: Medium
Note: You can exit anytime by typing 'exit'

What is the Vendor ID of the captured USB HID device?
Format: 0x1234 (or 1234)
> 0x046d

What is the Product ID of the captured USB HID device?
Format: 0x1234 (or 1234)
> 0xc31c

What is the USB device address assigned to the keyboard?
Format: int
> 7

What is the secret message decoded from the captured keystrokes?
Format: string
> wired_protocol_7_is_alive_2026
Congratulations! Here is your flag: KOMJAR26(USB_K3ystr0k3_zWmYRYEEQXKwDwQBtrYQomHf)
```

Langkah 3: Kirim ke Socket Server

Setelah mendapatkan Vendor ID, Product ID, Device Address, dan Pesan Rahasia, buka terminal node lab dan jalankan:

```
nc [IP_Group] 3402
```

Lalu masukkan jawaban sesuai format yang diminta oleh server interaktif tersebut.

16. 1. Identifikasi Alamat IP Server FTP Penyerang & Banner Software

- Buka file `wired_ftp_theft.pcap` menggunakan Wireshark.
- Filter lalu lintas FTP dengan mengetikkan `ftp` atau `ftp-data` pada kolom filter atas.
- Cari paket awal di mana koneksi TCP dibuat dan server merespons klien.
- Banner Software: Pada paket tanggapan pertama dari server (biasanya kode respons `220`), akan melihat teks banner selamat datang dari server FTP yang mencantumkan nama dan versi software (misalnya, `vsftpd` atau `ProFTPD` beserta versinya).
- IP Server: Alamat IP sumber dari paket yang mengirimkan banner 220 tersebut adalah alamat IP server FTP.

JAWABAN

Perhatikan paket nomor 64 (Source: `198.51.100.7`, Destination: `10.7.3.50`).

IP Server FTP: 198.51.100.7

Banner Software: Pada Info paket 64 tertulis:

Response: 220 Welcome to Wired FTP Server (vsftpd 3.0.5)

Jadi banner software-nya adalah vsftpd 3.0.5 (atau lengkapnya: Welcome to Wired FTP Server (vsftpd 3.0.5)).

2. Identifikasi Kredensial Login Penyerang

- Cari paket yang berisi perintah autentikasi FTP dalam bentuk teks jelas (cleartext):
 - USER [username]: Menunjukkan nama pengguna yang digunakan penyerang untuk masuk.
 - PASS [password]: Menunjukkan kata sandi yang dikirimkan setelahnya.
- Karena FTP str tidak mengenkripsi kredensial, username dan password akan terlihat secara langsung pada Packet Details (Transmission Control Protocol / File Transfer Protocol).

JAWABAN

Perhatikan paket setelahnya yang melibatkan IP 10.7.3.50 dan 198.51.100.7:

- User (Paket 66): USER knights_agent \$rightarrow\$ Username: knights_agent
- Pass (Paket 70): PASS N4v1_s3cur3_2026 \$rightarrow\$ Password: N4v1_s3cur3_2026

3. Identifikasi Ukuran File Malware (knights_payload.exe)

- Lanjutkan pencarian pada aliran (follow stream) atau filter percakapan FTP untuk melihat perintah transfer file, seperti RETR knights_payload.exe (mengunduh file).
- Untuk mengetahui ukuran byte secara persis:
 - Cari paket respons dari server yang mengonfirmasi transfer atau gunakan informasi dari aliran paket data FTP (ftp-data).
 - juga dapat melihat ringkasan paket transfer atau mengecek Packet Length dari segmen TCP yang membawa payload file tersebut, atau melihat detail respons perintah seperti 150 Opening BINARY mode

data connection hingga 226 Transfer complete yang sering kali menyertakan informasi ukuran file dalam byte.

JAWABAN

Perhatikan paket nomor 82 dan 84:

- *Paket 82: Request: SIZE knights_payload.exe (dari klien 10.7.3.50)*
- *Paket 84: Response: 213 524288 (dari server 198.51.100.7)*

Angka 213 adalah kode status FTP untuk file size, dan angka di belakangnya yaitu 524288 adalah ukuran file tersebut dalam bytes.

Jadi ukuran filenya adalah 524288 bytes.

Validasi Tean ke Socket Server

Setelah mendapatkan sea data yang diperlukan (IP server, banner, kredensial, dan ukuran file dalam byte), lakukan validasi otomatis atau interaktif ke server tantangan menggunakan perintah netcat di terminal:

nc [IP_Group] 3403

Masukkan tean sesuai format yang diminta oleh sistem server untuk menyelesaikan validasi soal ini.

RANGKUMAN JAWABAN

IP Server FTP: 198.51.100.7

Banner: vsftpd 3.0.5 (atau Welcome to Wired FTP Server (vsftpd 3.0.5))

Username: knights_agent

Password: N4v1_s3cur3_2026

File Size: 524288

flag: KOMJAR26{FTP_Th3ft_R2ezQCoEsXG66qwXkkMeQTanq}

```
alice-PUTTY
root@alice:~# nc 10.4.89.250 3403

===== Soal 16 - Protocol 7: FTP Credential Theft =====
Difficulty: Hard
Note: You can exit anytime by typing 'exit'

What is the IP address of the FTP server used to download the malware?
Format: IP
> 198.51.100.7

What FTP server software banner is returned upon connection?
Format: string
> vsftpd 3.0.5

What credential did the attacker use to log in to the FTP server?
Format: user:pass
> knights_agent
Wrong answer. Try again.

What credential did the attacker use to log in to the FTP server?
Format: user:pass
> knights_agent:N4v1_s3cur3_2026

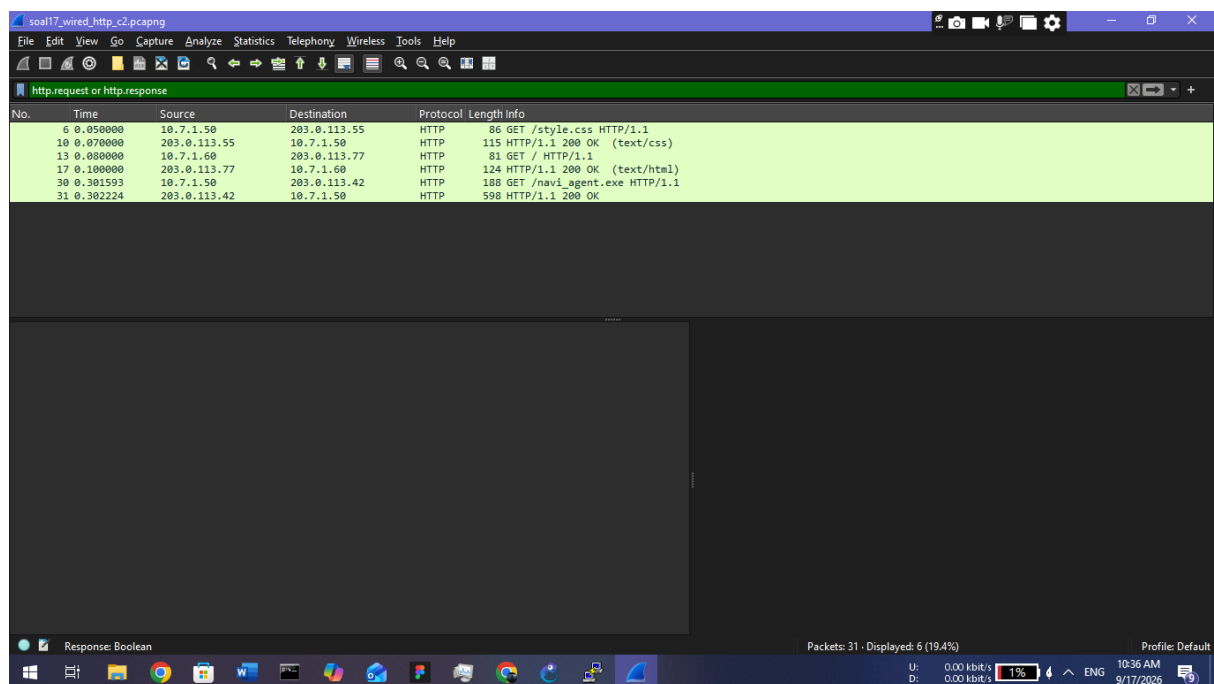
What is the size in bytes of the malware file (knights_payload.exe) requested via FTP?
Format: int
> 524288
Congratulations! Here is your flag: KOMJAR26{FTP_Th3ft_R2ezQCoEsXG66qXkkMeQTanq}
```

17. Buka file *wired_http_c2.pcap* di Wireshark.

Ketik filter di bagian atas dengan:

http.request or http.response

atau cukup ketik *http* untuk melihat seluruh percakapan HTTP.



Perhatikan paket nomor 30 (Metode **GET**):

Di kolom Info tertulis **GET /navi_agent.exe HTTP/1.1**.

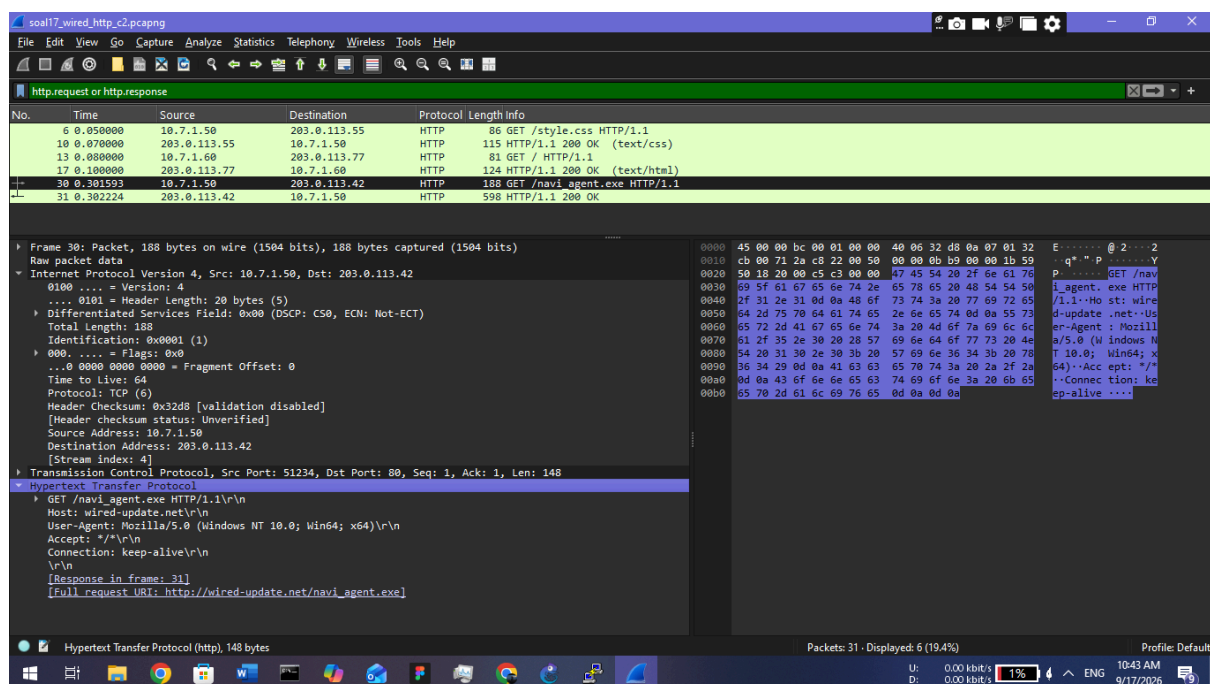
Nama file malware: **navi_agent.exe**

Alamat IP Server Penyerang (Destination dari paket 30 / Source dari paket 31):

IP Tujuan/Sumber pada transaksi tersebut adalah **203.0.113.42**.

Kode Status HTTP:

Perhatikan paket nomor 31 (respons dari server). Di kolom Info tertulis **HTTP/1.1 200 OK**. Jadi kode statusnya adalah **200** (atau **200 OK**).



Cari aktivitas unduhan file (biasanya metode **GET**):

- Nama Domain (Host): Lihat pada kolom Host atau Line-based text data di rincian paket **GET** untuk mengetahui nama domain tempat file tersebut diambil.
- Alamat IP Server Penyerang: Lihat pada Destination IP (atau Source IP pada respons server) dari server tempat file diunduh.
- Nama File Executable Malware: Cari jalur URL atau URI Path pada permintaan **GET** yang berakhir ekstensi program (seperti **.exe**).
- Kode Status HTTP: Lihat pada paket balasan dari server (biasanya paket **HTTP/1.1 200 OK** atau kode status lainnya) di kolom Info.

RANGKUMAN

Nama Domain / Host: **wired-update.net**

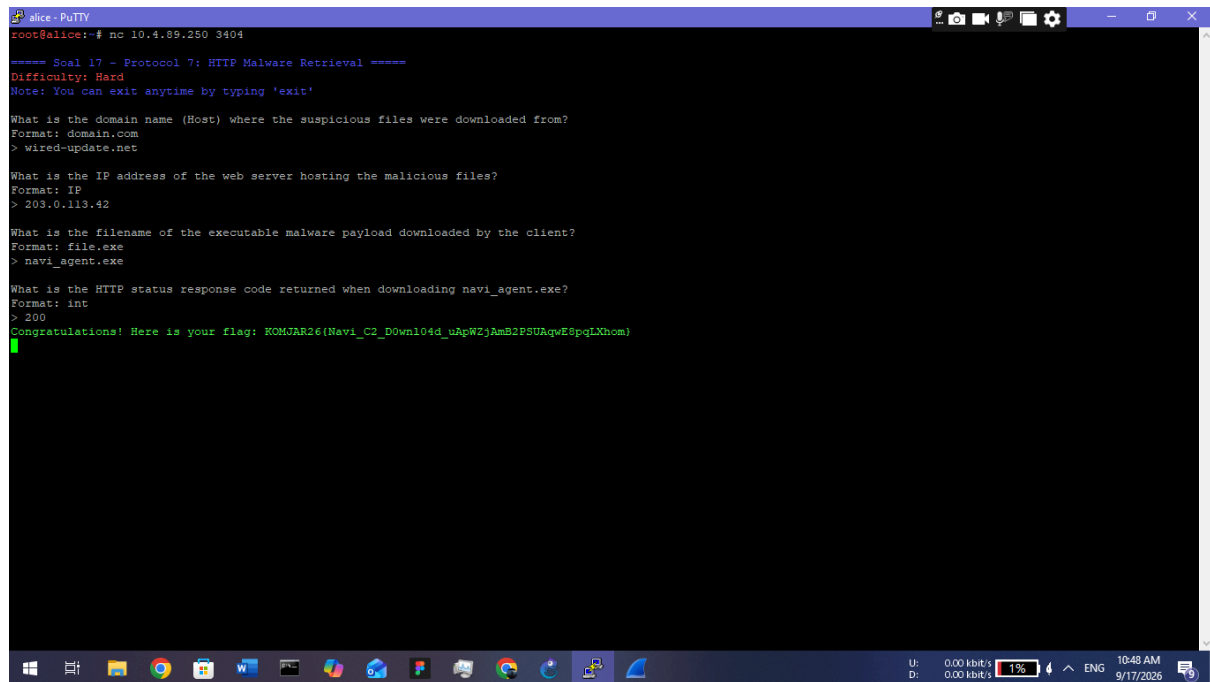
Alamat IP Server: 203.0.113.42

Nama File Malware: navi_agent.exe

Kode Status HTTP: 200 (atau 200 OK)

nc [IP_Group] 3404 flag:

KOMJAR26{Navi_C2_D0wnl04d_uApWZjAmB2PSUAqwE8pqLXhom}



```
alice-PuTTY
root@alice:~# nc 10.4.89.250 3404

===== Soal 17 - Protocol 7: HTTP Malware Retrieval =====
Difficulty: Hard
Note: You can exit anytime by typing 'exit'

What is the domain name (Host) where the suspicious files were downloaded from?
Format: domain.com
> wired-update.net

What is the IP address of the web server hosting the malicious files?
Format: IP
> 203.0.113.42

What is the filename of the executable malware payload downloaded by the client?
Format: file.exe
> navi_agent.exe

What is the HTTP status response code returned when downloading navi_agent.exe?
Format: int
> 200
Congratulations! Here is your flag: KOMJAR26{Navi_C2_D0wnl04d_uApWZjAmB2PSUAqwE8pqLXhom}
```

18. Buka file *wired_smb_transfer.pcapng* di Wireshark.

Ketik filter di bagian atas untuk menyaring lalu lintas SMB:

smb or smb2

Cari detail transaksi file sharing/transfer:

- Nama Protokol Jaringan yang Dieksploitasi: Secara spesifik adalah SMB (Server Message Block) atau SMBv2 / SMBv3.
- IP Pengirim dan Penerima: Perhatikan kolom Source dan Destination pada paket pertukaran file atau perintah buat/tulis file (Create Request, Write Request).
- Folder Tujuan Penyimpanan Malware (Share/Directory): Lihat pada jalur path direktori atau struktur tree connect pada detail paket SMB untuk mengetahui folder tempat file tersebut diletakkan korban (misalnya folder *shared*, *public*, *malware*, atau sejenisnya).

- Nama File Executable Malware: Cari nama file ber-ekstensi program (seperti `.exe`) yang dikirim atau dibuat melalui perintah SMB tersebut.

1. Nama Protokol Jaringan yang Dieksploitasi

- Terlihat pada kolom Protocol dan rincian paket, protokol yang digunakan adalah SMB2 (atau SMB).

2. IP Pengirim dan Penerima

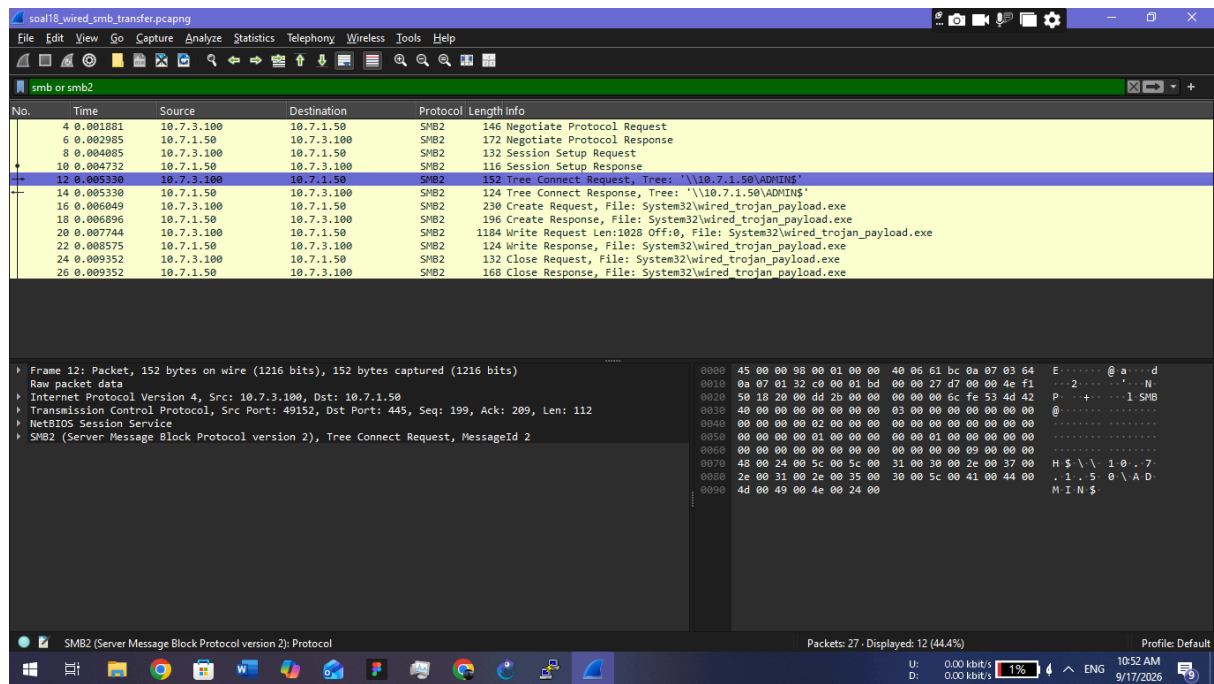
- IP Pengirim (Source): `10.7.3.100` (klien yang melakukan permintaan tulis/transfer file).
- IP Penerima (Destination): `10.7.1.50` (server korban/tujuan).

3. Folder Tujuan Penyimpanan Malware pada Sistem Korban

- Perhatikan paket nomor 12 (Tree Connect Request) atau paket-paket `Create Request` di bawahnya.
- Pada Info paket 12 tertulis: `Tree Connect Request, Tree: '\\10.7.1.50\ADMIN$'`.
- Selain itu, pada file `Create Request` (seperti paket 16), file diletakkan di dalam direktori `System32`.
- Jadi folder tujuannya adalah `ADMIN$` (atau path lengkapnya di folder `System32`).

4. Nama File Executable Malware yang Ditransfer

- Perhatikan kolom Info pada paket 16, 20, dan 24: `Create Request, File: System32\wired_trojan_payload.exe`
- Nama file malware yang ditransfer adalah `wired_trojan_payload.exe`.



RANGKUMAN

Protokol: **SMB2** (atau **SMB**)

IP Pengirim: **10.7.3.100**

IP Penerima: **10.7.1.50**

Folder Tujuan: **ADMIN\$ / System32**

Nama File Malware: **wired_trojan_payload.exe**

flag: **KOMJAR26{SMB_Tr4nsf3r_Tssoap3oiw7cU1I0Eq7f1dJSv}**

```
alice-PUTTY
root@alice:~# nc 10.4.89.250 3405

===== Soal 18 - Protocol 7: SMB Lateral Transfer =====
Difficulty: Hard
Note: You can exit anytime by typing 'exit'

What network file sharing protocol was used to transfer the malware to the victim?
Format: string
> SMB2

What is the IP address of the source host delivering the malware?
Format: IP
> 10.7.3.100

What is the IP address of the victim host receiving the malware?
Format: IP
> 10.7.1.50

What target share or directory on the victim was the malware written to?
Format: string
> ADMIN$ / System32
Wrong answer. Try again.

What target share or directory on the victim was the malware written to?
Format: string
> ADMIN$

What is the filename of the executable malware transferred?
Format: file.exe
> wired_trojan_payload.exe
Congratulations! Here is your flag: KCMJAR26(SMB_Tr4nsf3r_Tssoap3oiw7cUII0Eq7fidJSv)
```

19. Buka file *wired_smtp_threat.pcap* di Wireshark.

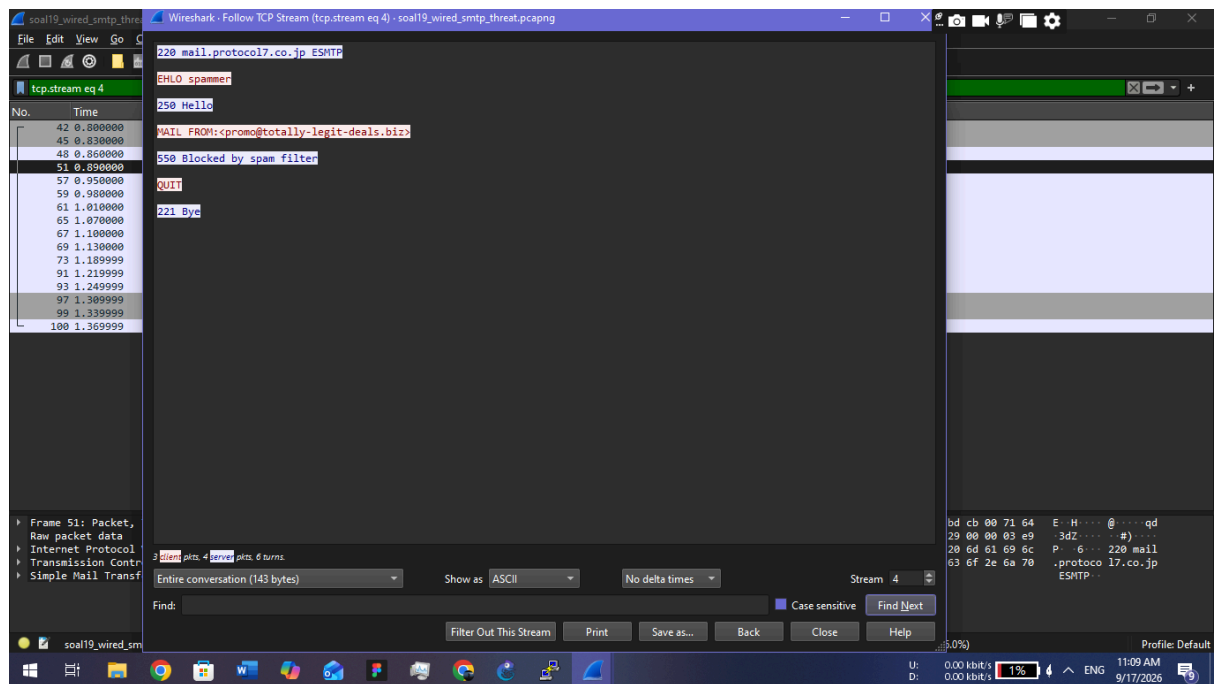
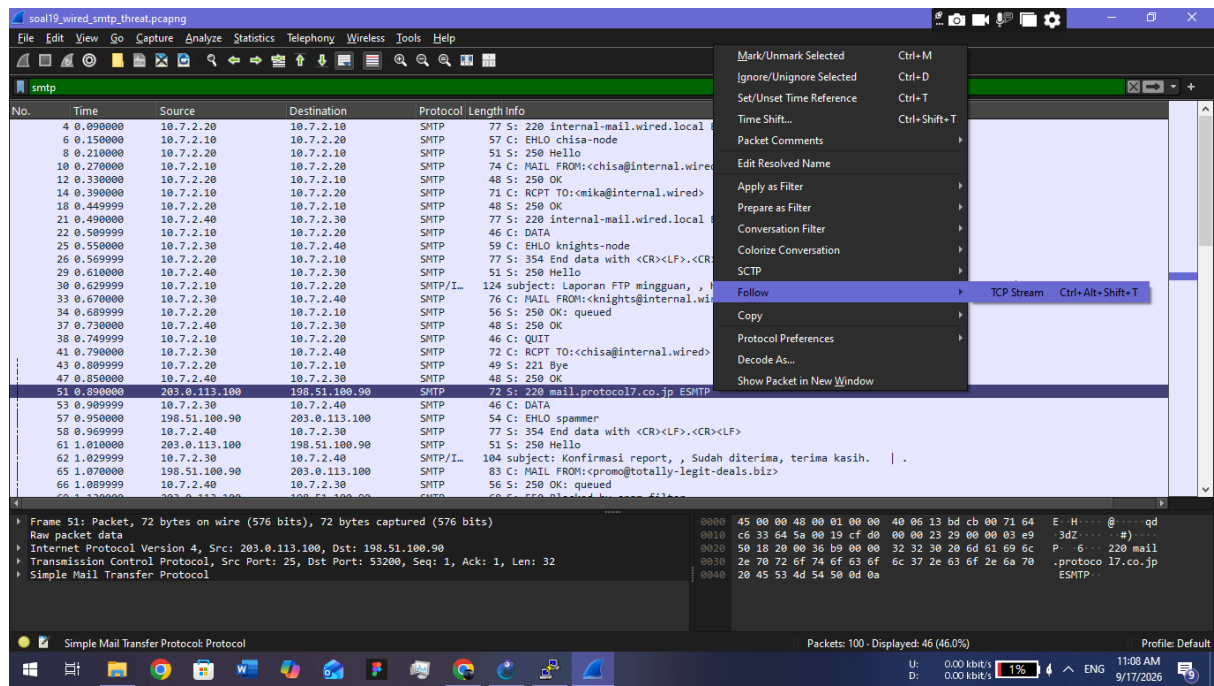
Ketik filter di bagian atas untuk menyaring lalu lintas SMTP atau TCP yang membawa data email:

smtp or tcp.port == 25

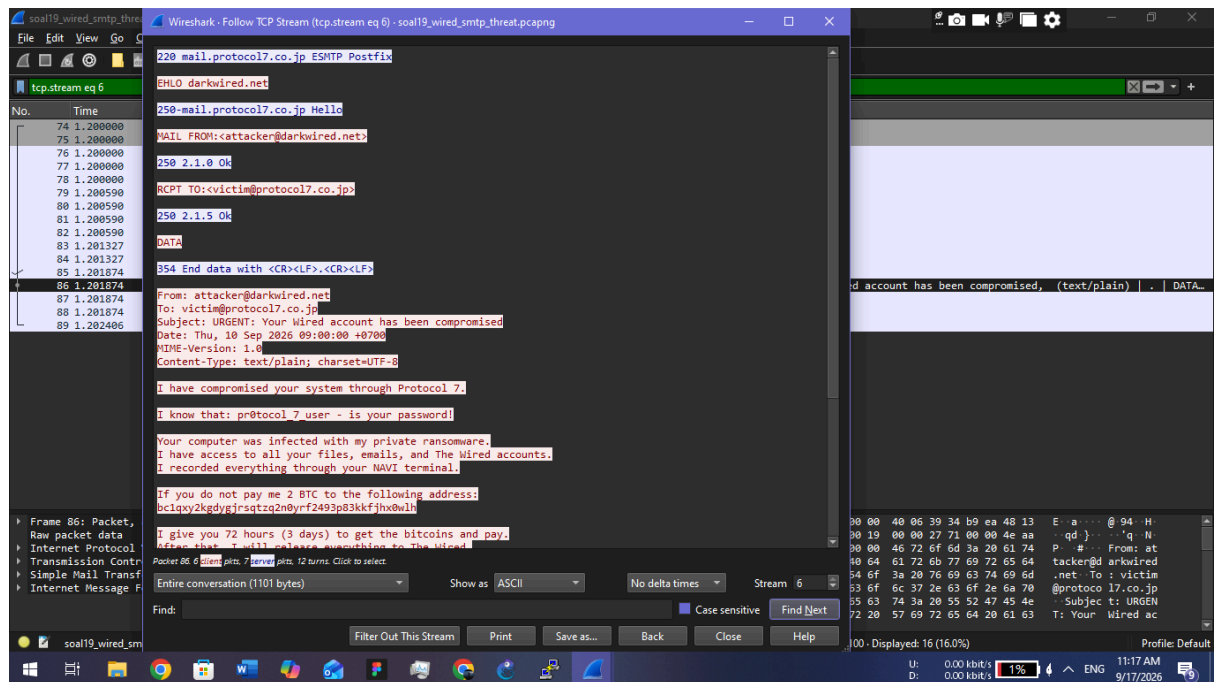
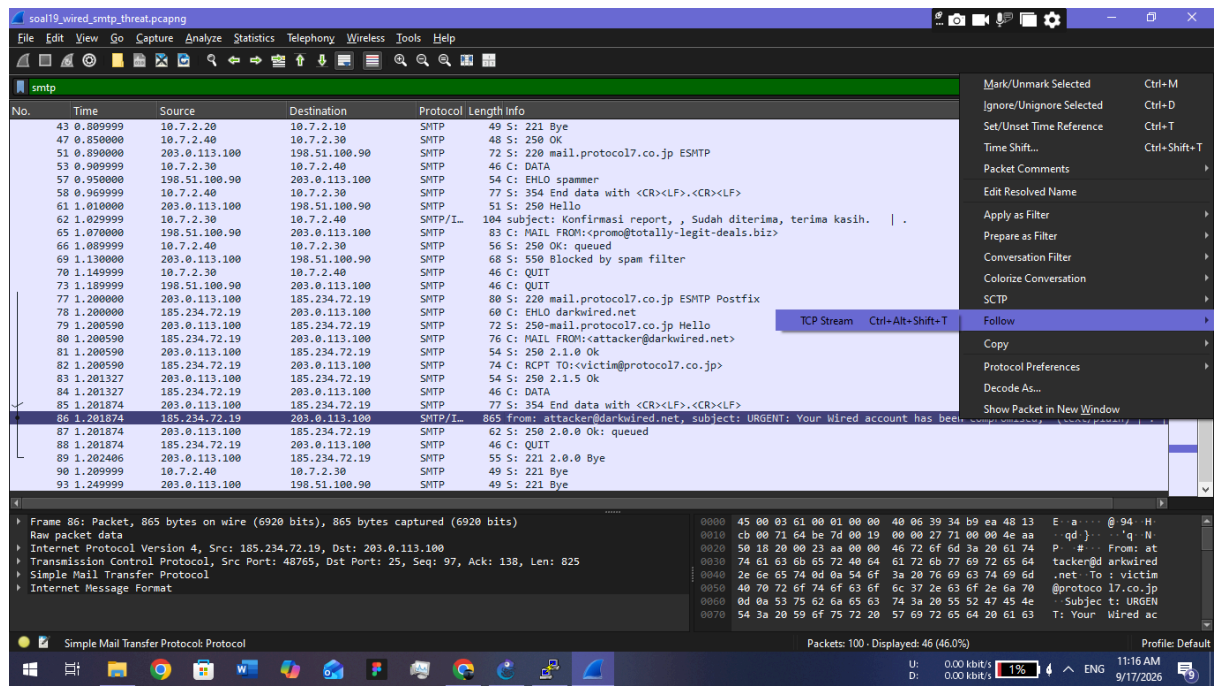
(Atau klik kanan pada salah satu paket SMTP, lalu pilih Follow > TCP Stream untuk membaca seluruh isi percakapan dari awal hingga akhir).

Cari poin-poin data yang diminta oleh soal pada teks percakapan (Stream):

- *Alamat Email Korban:* Cari pada baris tujuan penerima email (*RCPT TO:* atau bagian *To:* di dalam badan pesan).
- *Password Korban:* Cari teks di dalam badan email ancaman yang mengklaim kredensial/password korban telah bocor (biasanya dicantumkan sebagai bukti oleh pemeras).
- *Jenis Malware yang Diinfeksi:* Baca teks ancaman untuk mengetahui jenis malware yang diklaim telah disusupkan ke perangkat korban (misalnya Trojan, Spyware, Pegasus, RedLine, dll.).
- *Batas Waktu (dalam hari):* Cari angka durasi hari yang diberikan oleh penyerang untuk melakukan tebusan (misalnya 1 day, 2 days, 3 days, dll.).
- *MailClientID:* Cari string identifier atau MailClientID khusus yang tercantum di bagian footer/tangan pesan email tersebut.



Dua tangkapan layar terbaru (Screenshot 1213 dan 1214) menunjukkan bahwa kita sedang melihat *tcp.stream eq 4*, yang isinya diblokir oleh spam filter (550 Blocked by spam filter). Itu berarti ancaman tersebut ada di stream TCP nomor lain yang berhasil lolos.



Isi full email :

220 mail.protocol7.co.jp ESMTP Postfix

EHLO darkwired.net

250-mail.protocol7.co.jp Hello

MAIL FROM:<attacker@darkwired.net>

250 2.1.0 Ok

RCPT TO:<victim@protocol7.co.jp>

250 2.1.5 Ok

DATA

354 End data with <CR><LF>.<CR><LF>

From: attacker@darkwired.net

To: victim@protocol7.co.jp

Subject: URGENT: Your Wired account has been compromised

Date: Thu, 10 Sep 2026 09:00:00 +0700

MIME-Version: 1.0

Content-Type: text/plain; charset=UTF-8

I have compromised your system through Protocol 7.

I know that: pr0tocol_7_user - is your password!

Your computer was infected with my private ransomware.

I have access to all your files, emails, and The Wired accounts.

I recorded everything through your NAVI terminal.

If you do not pay me 2 BTC to the following address:

bc1qxy2kgdygjrqtzq2n0yrf2493p83kkfjhx0wlh

I give you 72 hours (3 days) to get the bitcoins and pay.

After that, I will release everything to The Wired.

Do not try to contact the Knights. They cannot help you.

Let's all love Lain.

MailClientID: 7719980706

.

250 2.0.0 Ok: queued

QUIT

221 2.0.0 Bye

RANGKUMAN

Alamat Email Korban (Targeted Email): victim@protocol7.co.jp (terlihat pada baris RCPT TO:<victim@protocol7.co.jp> / To:).

Password Korban yang Bocor: pr0tocol_7_user (terlihat pada kalimat "I know that: pr0tocol_7_user - is your password!").

Jenis Malware yang Diinfeksi: ransomware (atau private ransomware).

Batas Waktu (Deadline dalam hari): 3 (terlihat pada kalimat "72 hours (3 days)").

MailClientID: 7719980706 (terlihat pada bagian bawah baris MailClientID: 7719980706).

flag: KOMJAR26{SMTP_Ext0rt10n_ZFLfUJOsNM9wPysEm9m3ZlIt}

```
Alice-PUTTY
root@alice:~# nc 10.4.89.250 3406

===== Soal 19 - Protocol 7: SMTP Threat Inspection =====
Difficulty: Hard
Note: You can exit anytime by typing 'exit'

What is the email address of the victim targeted by the extortionist?
Format: user@domain.com
> victim@protocol7.co.jp

What password did the extortionist claim was stolen from the victim?
Format: string
> protocol_7_user

What type of malware did the attacker claim infected the victim's computer?
Format: string
> ransomware

How many days deadline did the attacker give the victim to pay?
Format: int
> 3

What is the MailClientID specified at the bottom of the extortion email?
Format: int
> 7719980706
Congratulations! Here is your flag: KOMJAR26(SMTP_Ext0rt10n_ZFLfjUJOsNM9wPysEm9m3ZlIt)
```

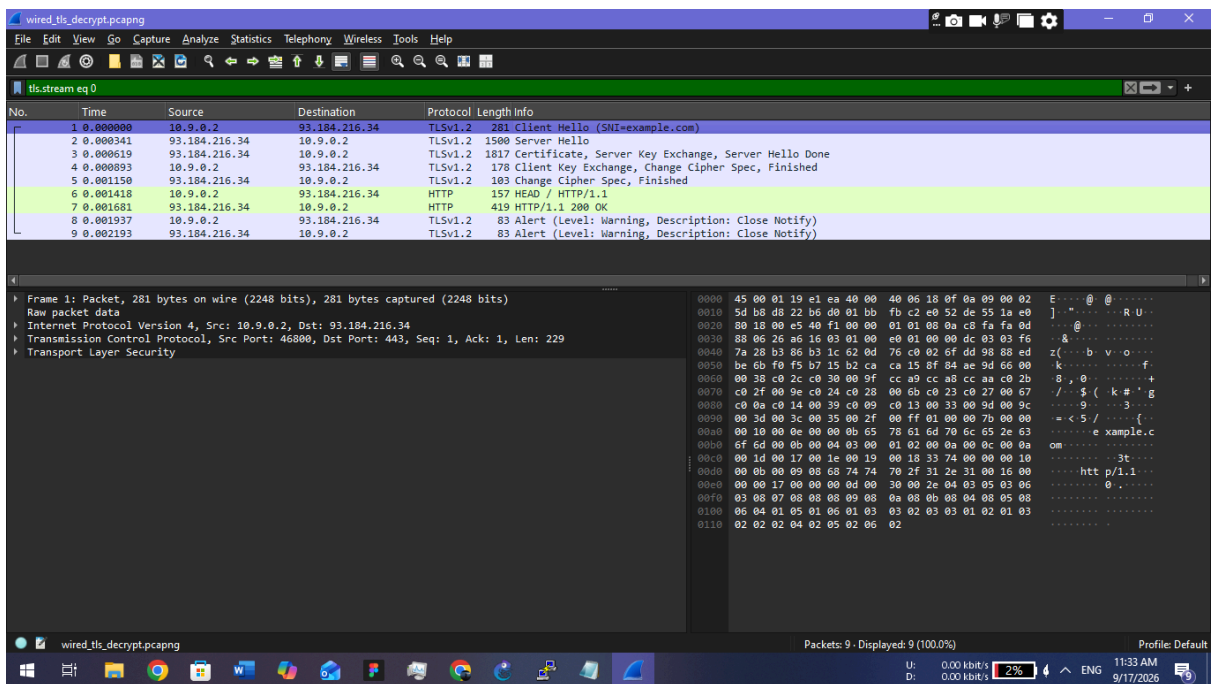
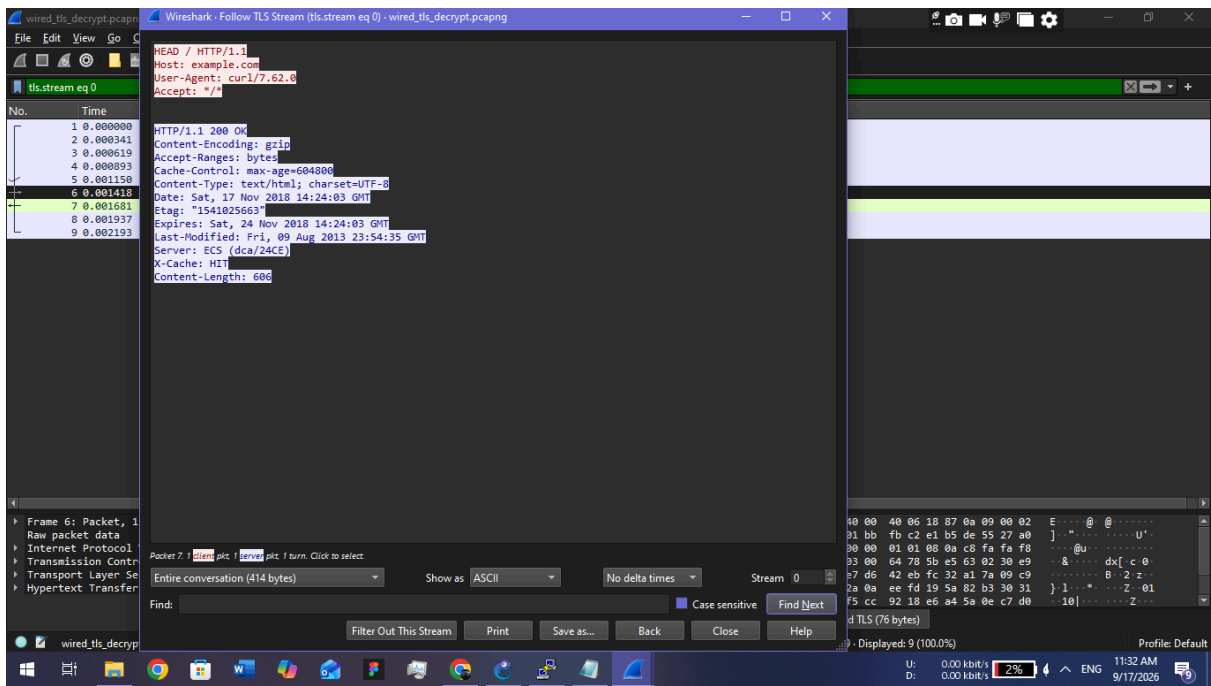
20. Buka file `wired_tls_decrypt.pcapng` di Wireshark.

1. Di menu atas, pilih `Edit > Preferences`.
2. Di jendela pengaturan sebelah kiri, luaskan menu `Protocols`, lalu klik `TLS` (atau `SSL` pada versi Wireshark lama).
3. Pada kolom (Pre)-Master-Secret log filename, klik tombol `Browse` (atau `Folder`) dan pilih file `keylogfile.txt` yang sudah berisi baris `CLIENT_RANDOM ...` tersebut.
4. Klik `OK / Save`.

Cara Menekan Data untuk Port 3407:

Setelah didekripsi, ketik filter di bagian atas Wireshark:

`http or tls`



RANGKUMAN

Versi Protokol TLS: **TLSv1.2** (terlihat pada Info paket nomor 1 dan rincian Client Hello).

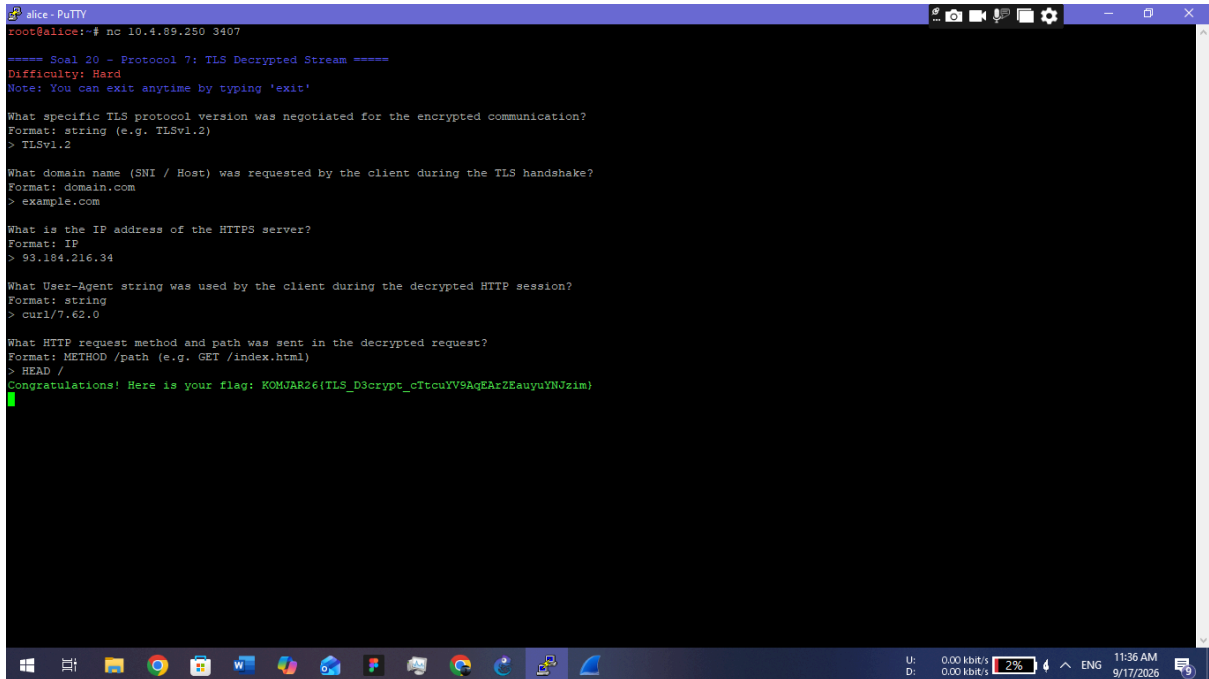
Nama Domain (SNI): **example.com** (terlihat langsung pada Info paket 1: **Client Hello** (SNI=example.com)).

Alamat IP Server HTTPS Penyerang: **93.184.216.34** (terlihat pada kolom Destination di paket 1 / Source IP dari server).

User-Agent: **curl/7.62.0** (seperti yang terlihat pada header HTTP di tangkapan layar sebelumnya).

HTTP Request Method & Path: **HEAD** dan path **/** (atau **HEAD /**).

flag: **KOMJAR26{TLS_D3crypt_cTtcuYV9AqEArZEauyuYNJzim}**



```
Alice-PuTTY
root@alice:~# nc 10.4.89.250 3407

===== Soal 20 - Protocol 7: TLS Decrypted Stream =====
Difficulty: Hard
Note: You can exit anytime by typing 'exit'

What specific TLS protocol version was negotiated for the encrypted communication?
Format: string (e.g. TLSv1.2)
> TLSv1.2

What domain name (SNI / Host) was requested by the client during the TLS handshake?
Format: domain.com
> example.com

What is the IP address of the HTTPS server?
Format: IP
> 93.184.216.34

What User-Agent string was used by the client during the decrypted HTTP session?
Format: string
> curl/7.62.0

What HTTP request method and path was sent in the decrypted request?
Format: METHOD /path (e.g. GET /index.html)
> HEAD /

Congratulations! Here is your flag: KOMJAR26{TLS_D3crypt_cTtcuYV9AqEArZEauyuYNJzim}
```